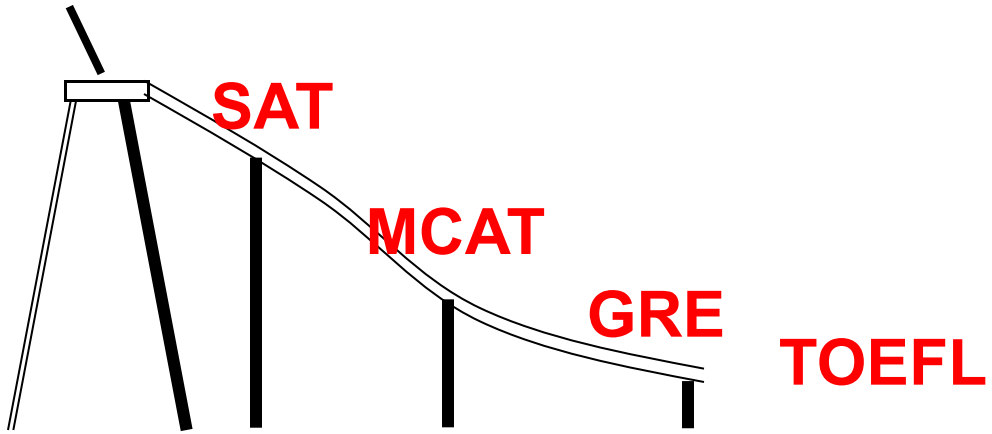
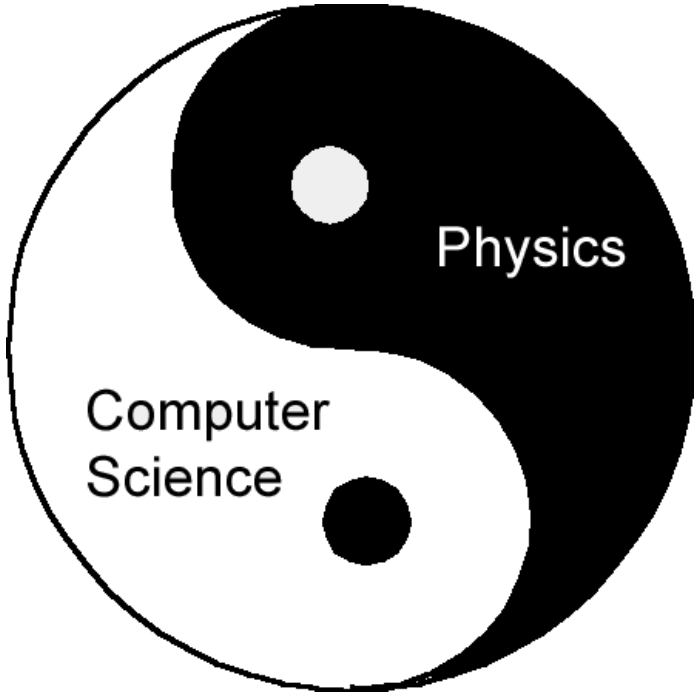
Test Slide



Tales from the Boundary of Physics and Computer Science

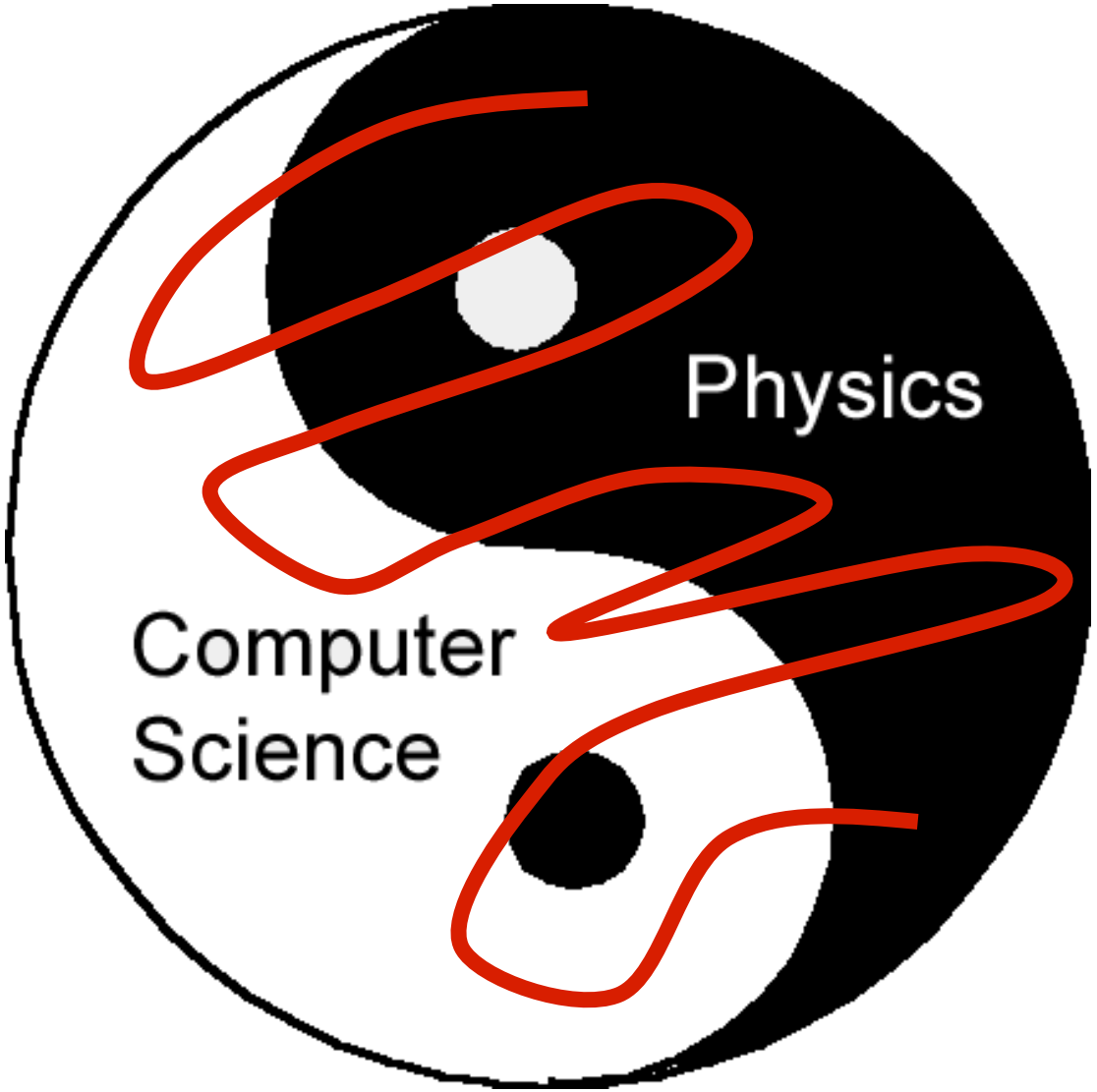


Dave Bacon
Caltech
Department of Physics
Institute for Quantum
Information

Mountains



Talk Outline



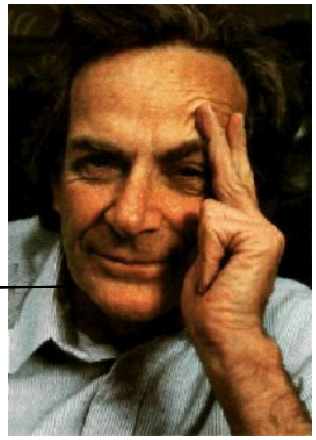
In the Beginning



Rolf Landauer
(IBM)

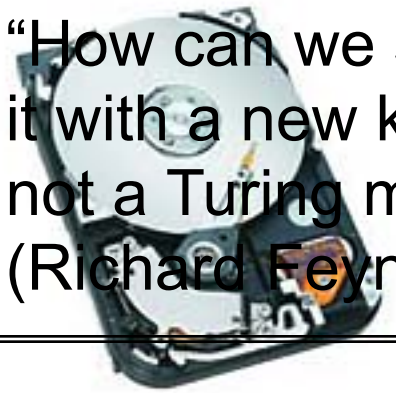
Err..I mean: Information carriers obey physics!

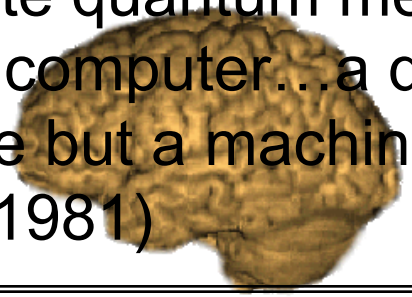
Physics is quantum!



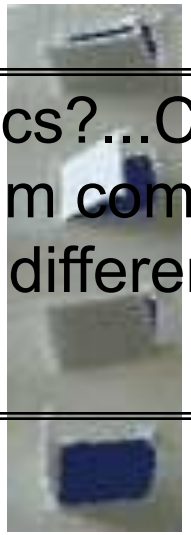
Richard Feynman

Information is “fungible”:





“How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It’s not a Turing machine but a machine of a different kind” (Richard Feynman, 1981)





Al Gore

Not All Computer's Are Equal

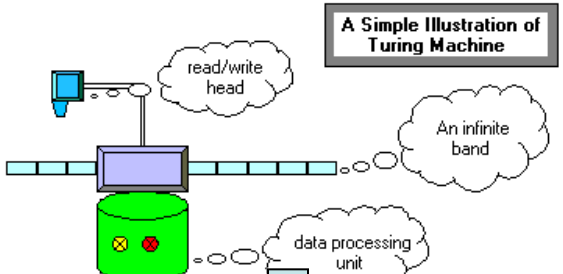
(apologies to Turing)

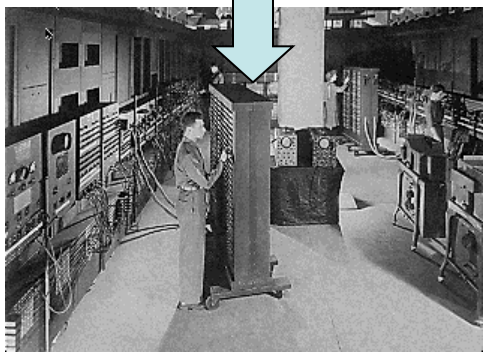
Where are all the quantum computers?



Alan Turing

"On Computable Numbers with an Application to the Entscheidungsproblem." (1938)



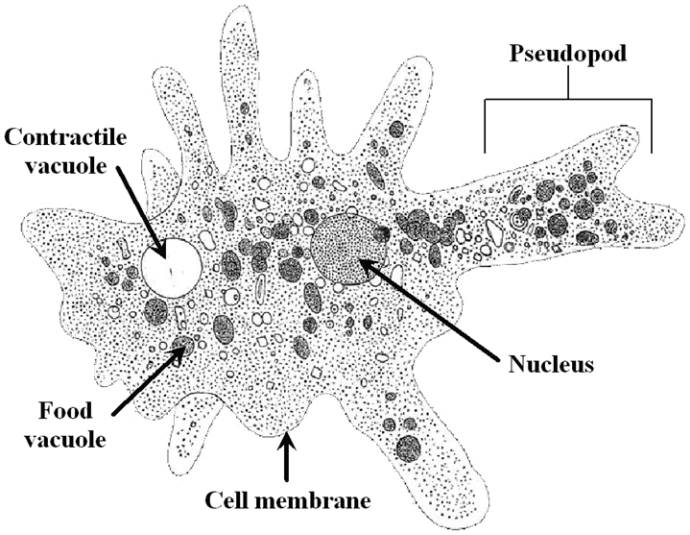


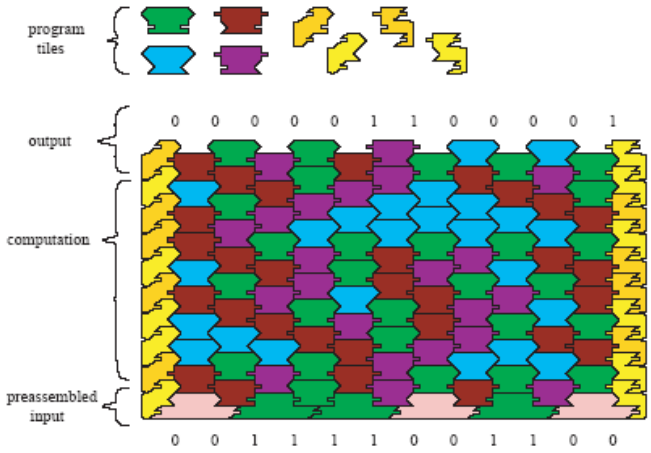
18000 vacuum tubes

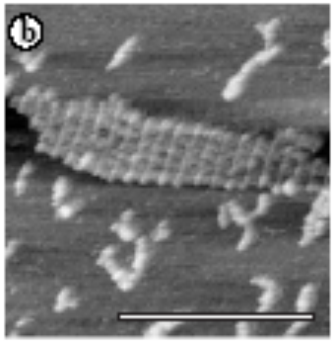
Failure rate:
50 tubes / day

Electronic Numerical Integrator and Computer (ENIAC, 1943)

Is it a ROBOT? Is it a Computer?





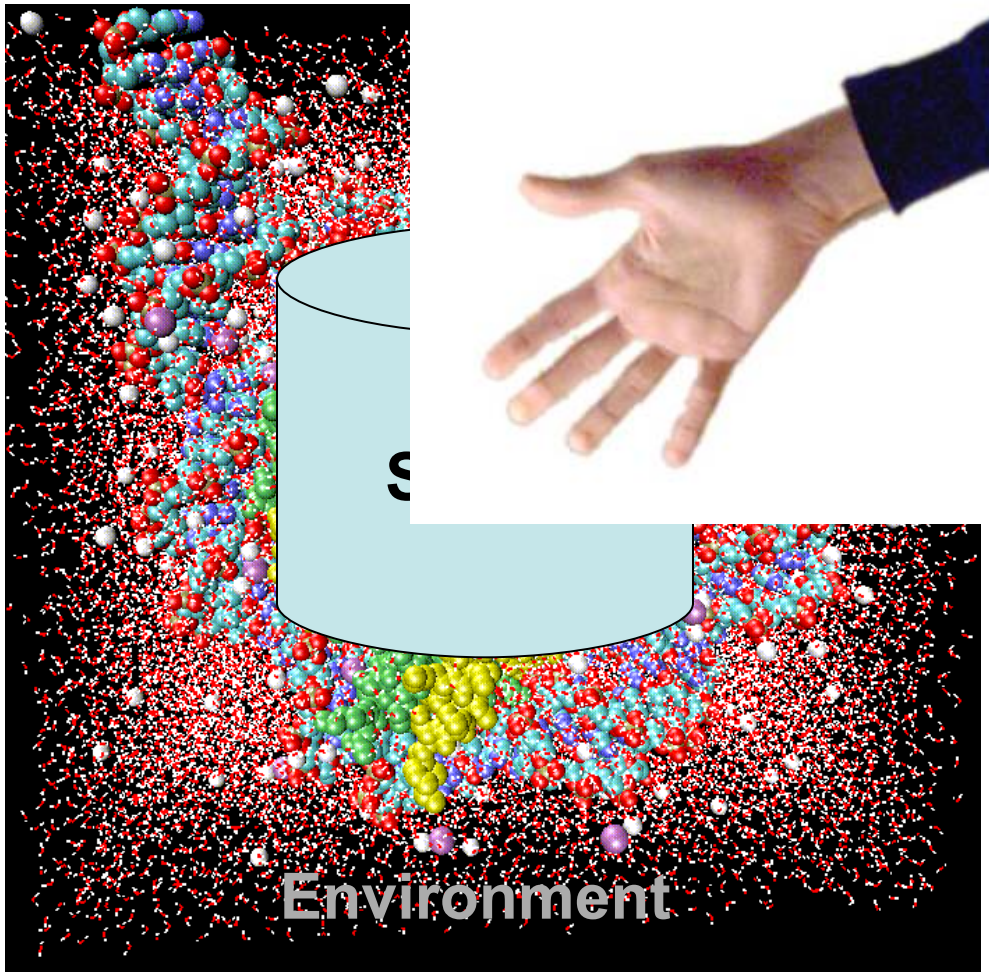


Universal DNA Computing

(pics from Eric Winfree's Caltech lab)

The Robust Computer Phase

A phase of matter?





Control

- Controllable yet Decoupled
- Switching, Digital
- Regular yet Complex

Reducible yet Irreducible

Computerology

The Answer Key

Theory:

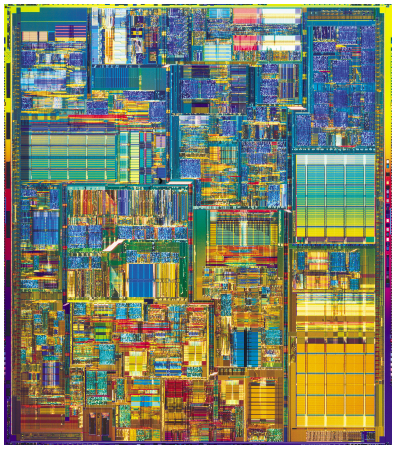
"A Mathematical Theory of Communication" - Shannon (1948)

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" - von Neumann (1956)

If control is precise enough and noise is small enough, robust classical computation is possible

Practice:

Integrated Circuit



Hard Drive



Rant mode ON

The Physics Guarantee

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation:

there exists systems whose <u>PHYSICS</u> guarantees their ability to enact robust classical computation.

Two Paths

Hard Drive

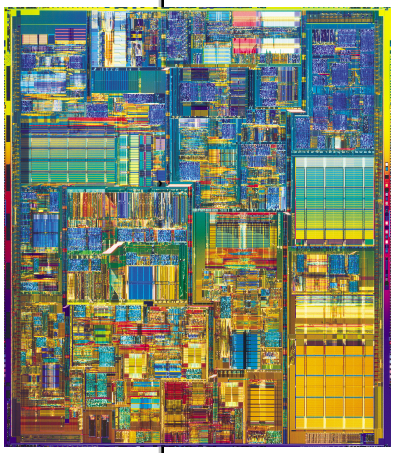


Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

Integrated Circuit



Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

Rant mode ON

The Physics Guarantee

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

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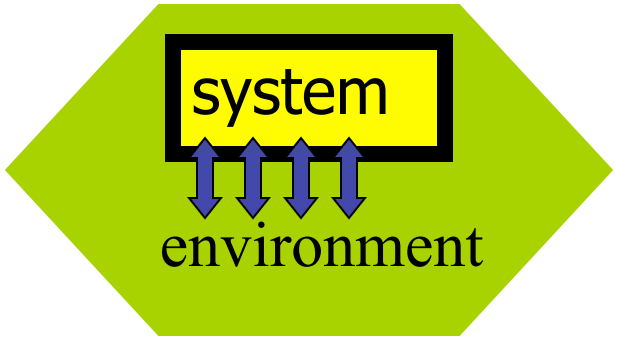
<u>THE BILLION DOLLAR QUANTUM QUESTION:</u>

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

What is the Problem?

Where are all the quantum computers?

Real quantum systems are open quantum systems!



Full Hilbert space $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$

Evolution is $\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

System evolution — environment evolution — **Interaction causing DECOHERENCE**

Hamiltonian:

$$\mathbf{H}_{SB} = \mathbf{H}_S \otimes \mathbf{I}_B + \mathbf{I}_S \otimes \mathbf{H}_B + \mathbf{H}_I$$

System's pure states become entangled with environment:

$$|\psi_0\rangle = |\psi\rangle_S \otimes |\psi\rangle_B \rightarrow \sum_n c_n |n\rangle \otimes |n\rangle \qquad \rho_0 = |\psi_0\rangle_S \, {}_S\langle\psi_0| \rightarrow \text{mixed state}$$

System evolution (by itself) is no longer unitary

Coherent properties of system's quantum information lost

Problem for quantum computers: [Unruh, *Phys. Rev. A* 51, 992 (1995)]

What Is The Problem?

Decoherence is a lot like classical noise, BUT:

Strong coupling to control devices needed to enact computations



Strong coupling to environment causes decoherence

Yingyang of quantum computing

Quantum Computing is Bunk

(1994 edition)

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$$

Measurements are faulty

measurement result is noisy, incorrect result obtained

Quantum Computing Disappearing Act

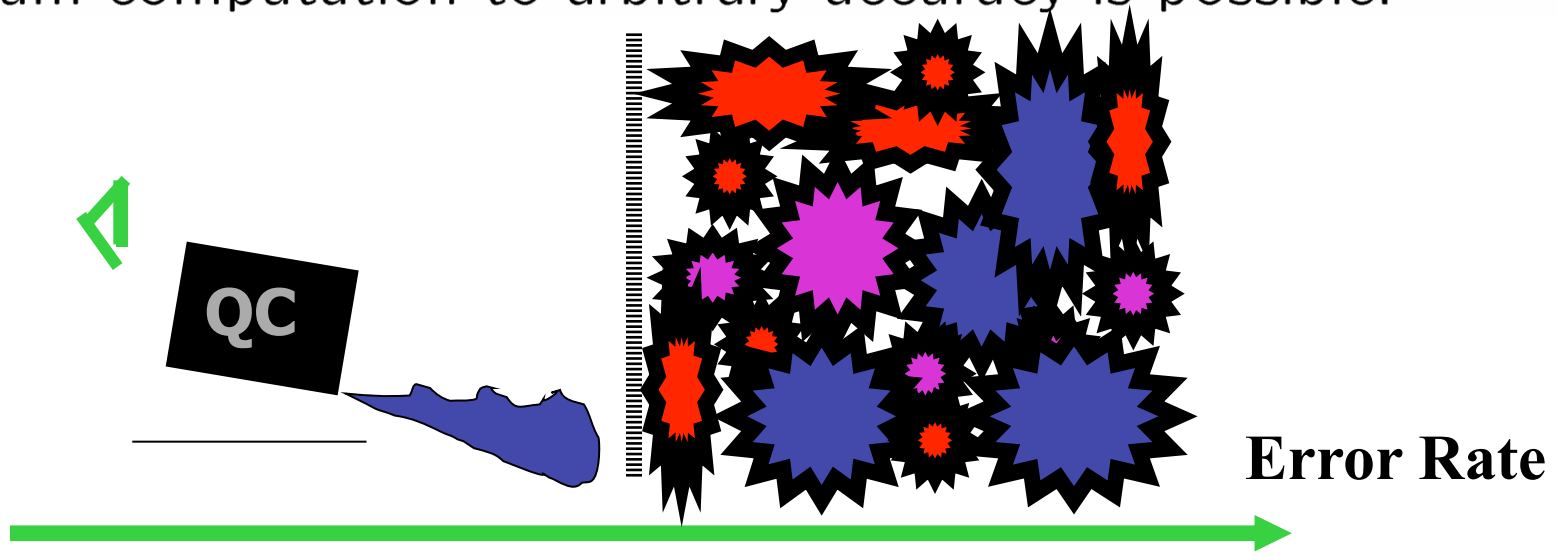
qubits disappear (leakage of computing states)

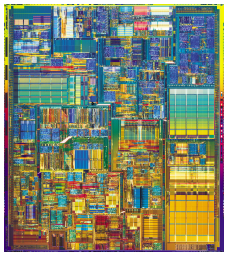
The Quantum Solution (1995-96)

Fault Tolerant Quantum Computing is Possible!

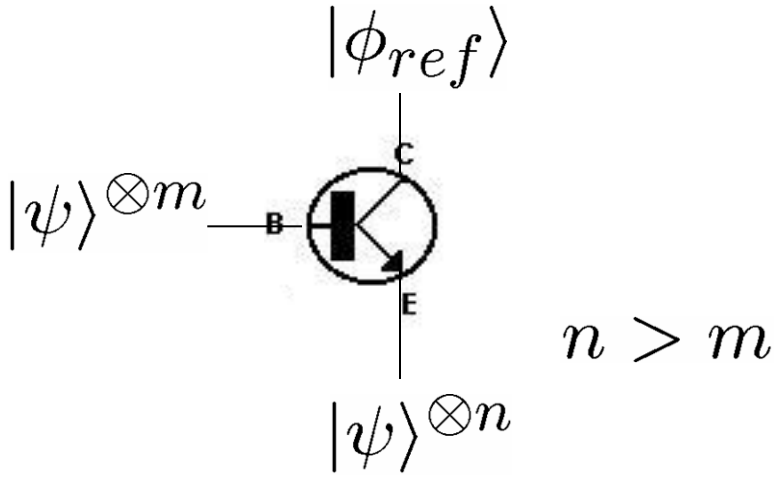
Threshold

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.



BUT:  ,  ?

No Quantum Transistor





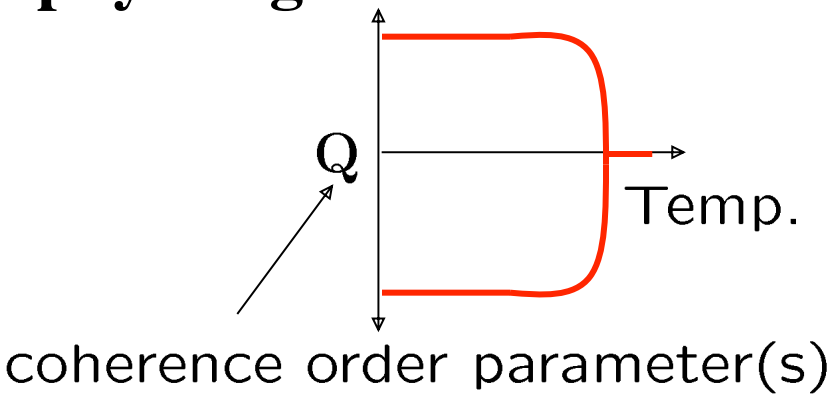
Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

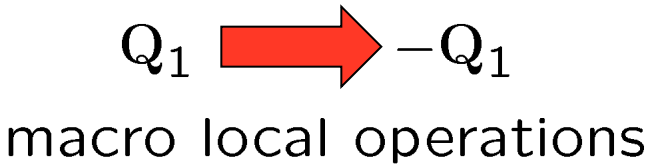
The Quantum Hard Drive?

[Bacon, Ph.D. Thesis, U.C. Berkeley Physics "Decoherence, Control, and Symmetry in Quantum Computers," (2001)]

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.



2. Accessible Fault-Tolerant Operations

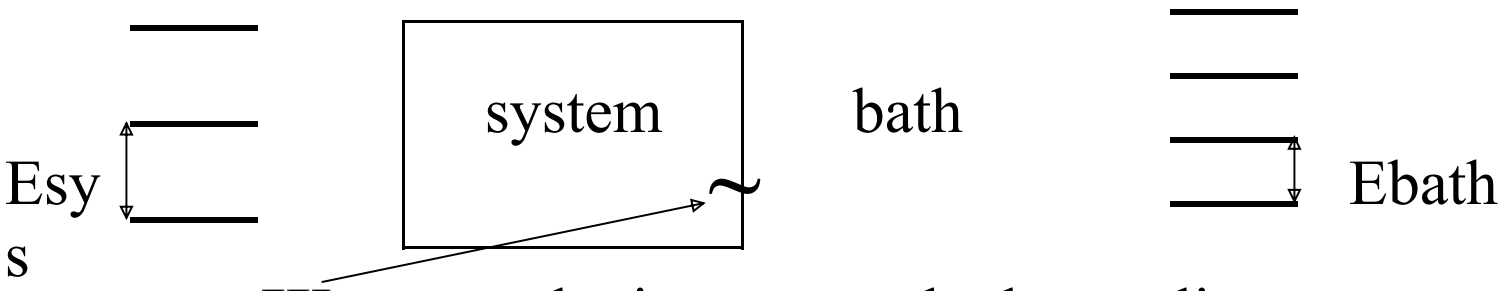
"self-correcting"

$$\left\langle G_1, \ldots, G_{n^k} \right\rangle \cong U(c^n)$$

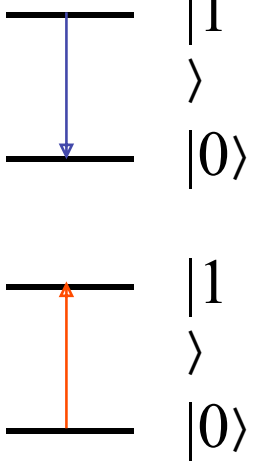
3. Universality

First Steps Towards Using Physics

[Bacon, Brown, and Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)]



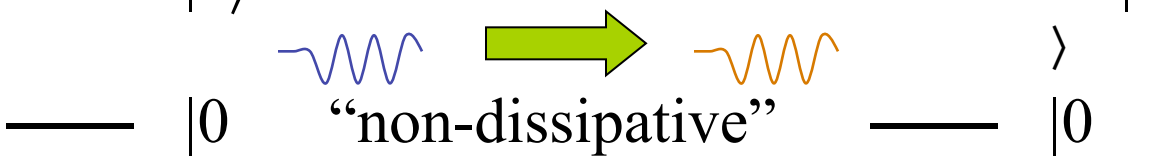
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies*



$$\mathbf{H}_I = \eta \sum_k g_k (\sigma_+ + \sigma_-)(\mathbf{a}_k + \mathbf{a}_k^\dagger)$$

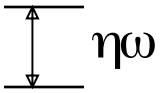
$$\not\approx \quad \eta \sum_k g_k (\sigma_+ \mathbf{a}_k + \sigma_- \mathbf{a}_k^\dagger)$$

rotating wave approximation



[e.g. Prokof'ev and Stamp, *J. Phys. Cond. Mat.* 5, 663

two-level atom radiatively coupled to a thermal reservoir:



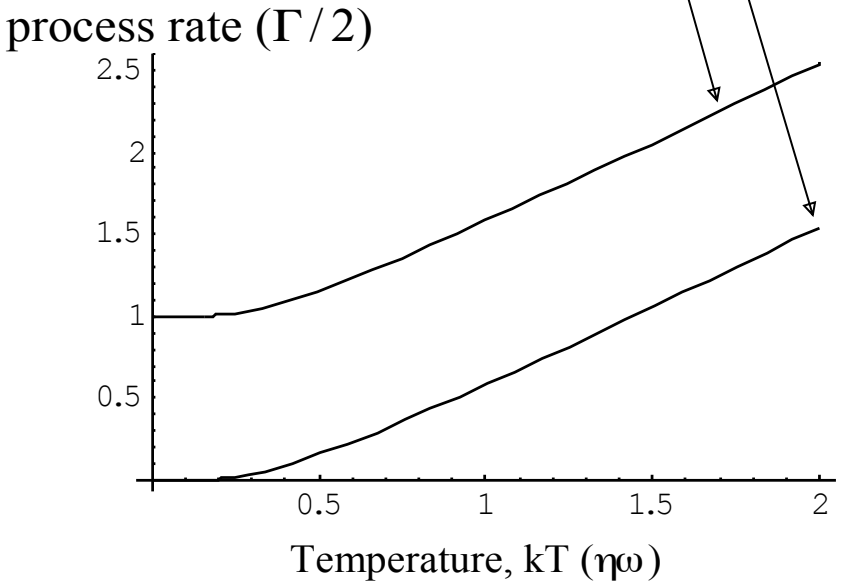
$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\Gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right] \quad \text{“heating”}$$

$$-(\bar{n}_{th}+1)\frac{\Gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right] \quad \text{“cooling”}$$

$$\left\{ \varphi\left[\rho - \sigma_z\rho\sigma_z\right] \right. \quad \text{“non-dissipative”}$$

non-dissipative dephasing

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



At low bath temperatures, heating disappears……..

Can We Make Quantum Errors Run Uphill?

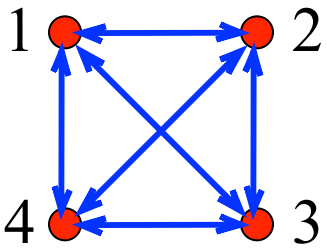
A Supercoherent Qubit

[Bacon, Brown, and Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)]

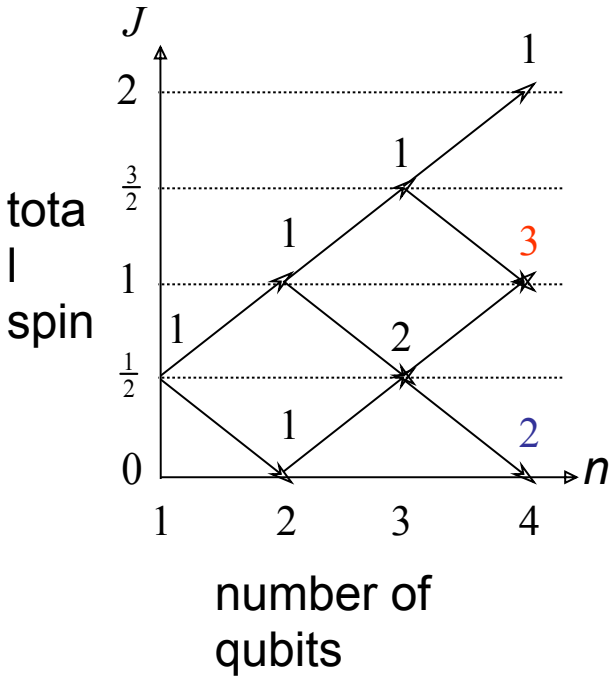
[Bacon, U.C. Berkeley Ph.D. Thesis (2001)]

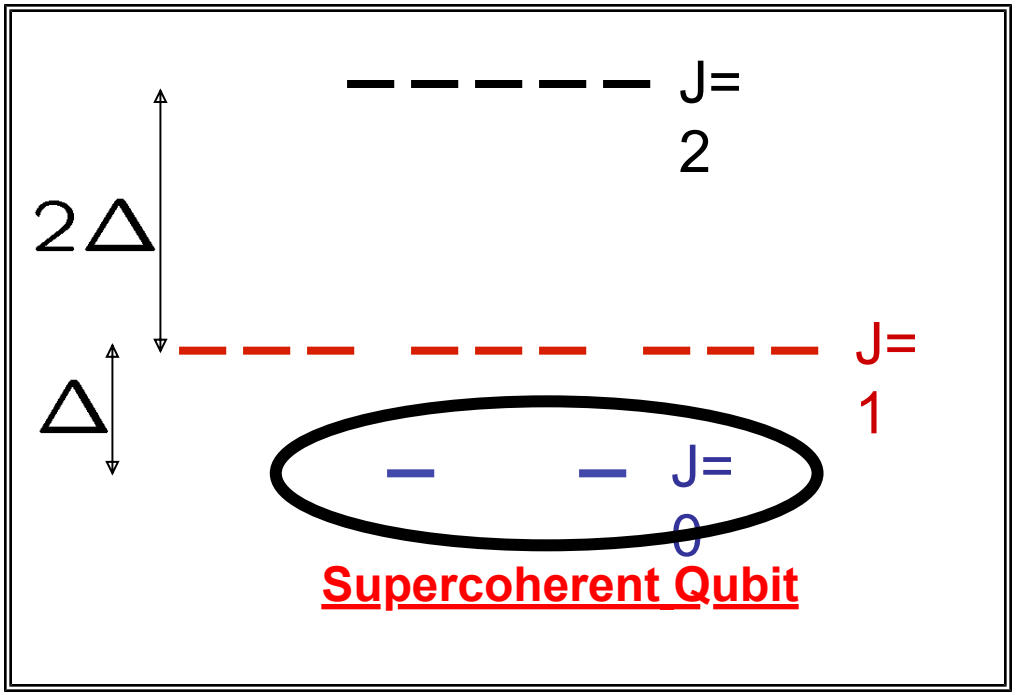
$$\mathrm{H} = \Delta \sum_{i>j=1}^{4} \hat{\mathrm{s}}_i \cdot \hat{\mathrm{s}}_j = \frac{\Delta}{2}\left(\hat{\mathrm{S}}^2 - 3\mathrm{I}\right)$$

$$\hat{s}_i = (\mathrm{s}_x, \mathrm{s}_y, \mathrm{s}_z)|_{i\text{th qubit}}$$

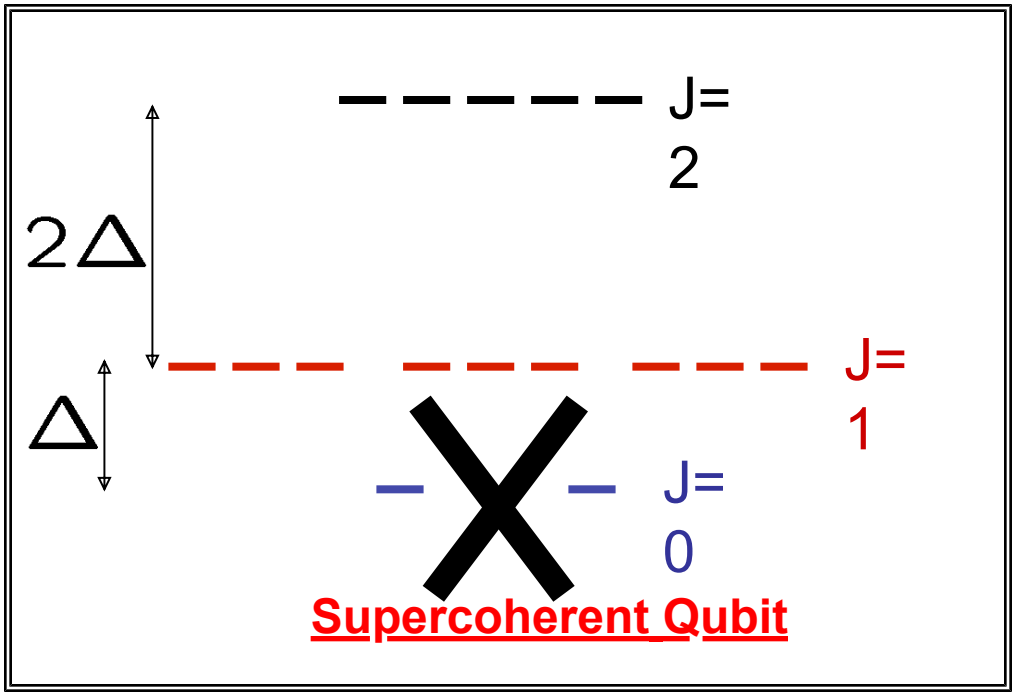


↔ = exchange interaction





Run Uphill

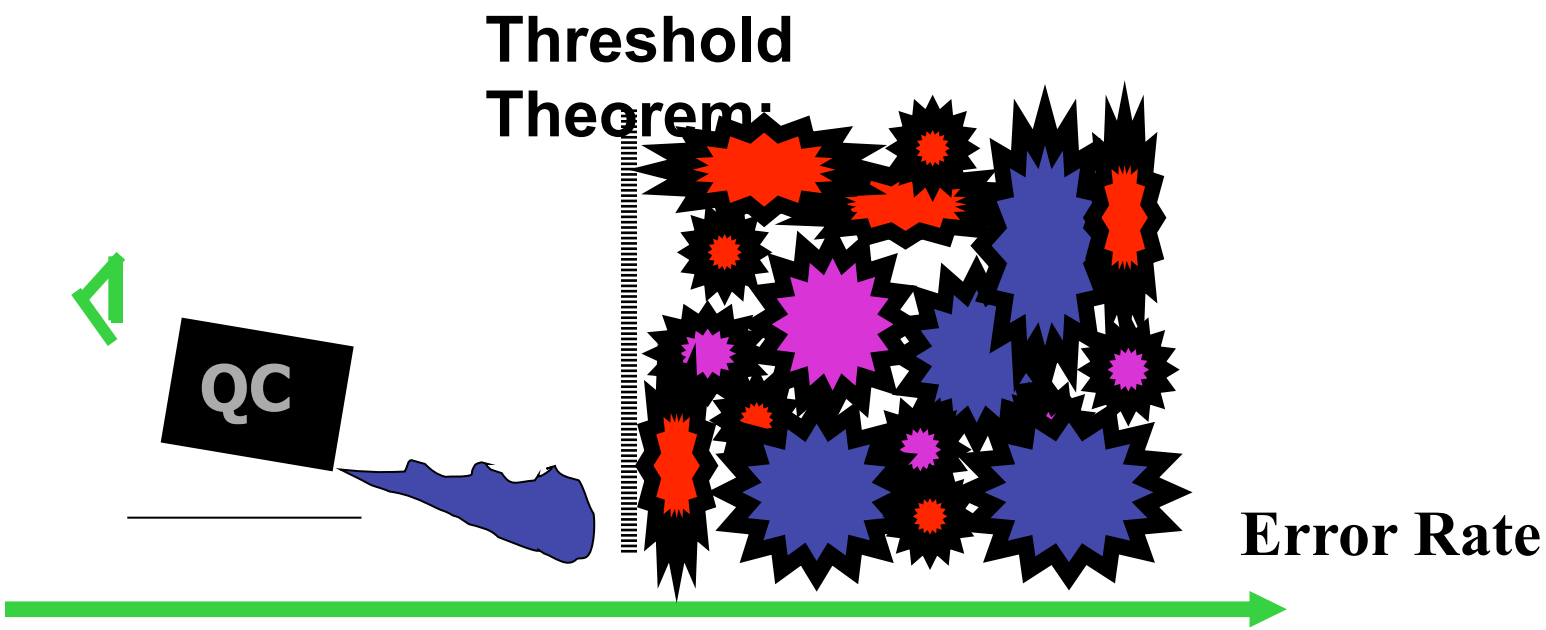


All single qubit operators $(s_i)_\alpha$ take J=0 states to J>0 states

Supercoherence:
made all single qubit prosses "heating"!

At kT< Δ single qubit decoherence mechanisms suppressed

Quantum Error Correction



Classical code

$$0 \rightarrow 000$$
$$1 \rightarrow 111$$

Probability of

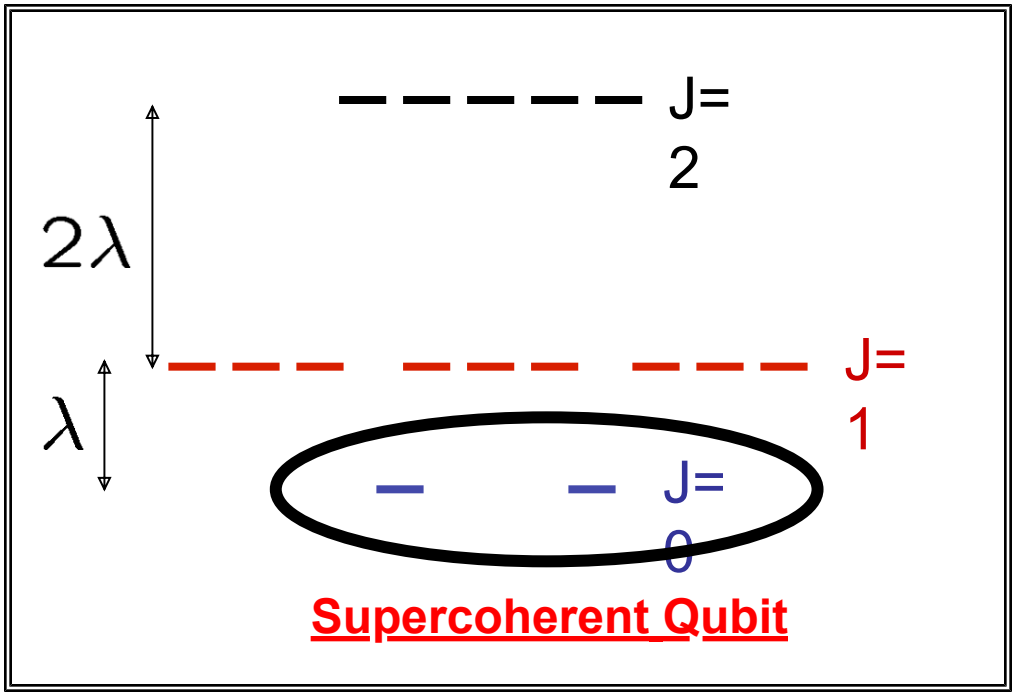
$$= 3p^2(1-p) + p^3$$
$$< p \text{ if } p < \tfrac{1}{2}$$

Quantum code

$$|0\rangle \rightarrow |\psi_0\rangle$$
$$|1\rangle \rightarrow |\psi_1\rangle$$

qubit encoded into quantum error
correcting subspace

There's a Code in My Hamiltonian



classical error detecting code:

$$0 \rightarrow 00$$
$$1 \rightarrow 11$$

the degenerate ground state = quantum error detecting code

supercoherent qubit code: $\langle i_L|(\mathbf{s}_i)_\alpha|j_L\rangle = 0$

$$|0_L\rangle \rightarrow \tfrac{1}{2}(|01\rangle - |10\rangle)(|01\rangle - |10\rangle)$$

$$|1_L\rangle \rightarrow \frac{1}{\sqrt{12}}(2|0011\rangle + 2|1100\rangle - |0101\rangle - |1010\rangle - |1001\rangle - |0110\rangle)$$

Decoherence-Free Subspaces

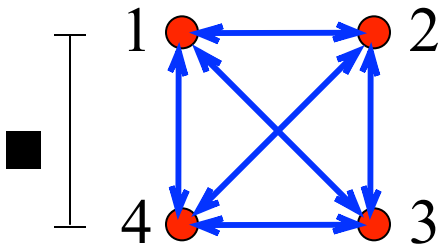
[Zanardi and Rasetti, *Phys. Rev. Lett.* 79, 1953 (1997)]

[Lidar, Bacon, and Whaley, *Phys. Rev. Lett.* 82, 4556 (1999)]

[Bacon, Kempe, Lidar, and Whaley, *Phys. Rev. Lett.* 85, 1758 (2000)]

[Kempe, Bacon, Lidar, and Whaley, *Phys. Rev. A* 63, 042307 (2001)]

[Bacon, U.C. Berkeley Ph.D. Thesis (2001)]



λ bath mode

$$H_I = \sum_{i=1}^{n} \sum_{\alpha=\{x,y,z\}} c_{\alpha,i} \; s_\alpha^{(i)} \otimes b_{\alpha,i} \quad \Leftrightarrow \quad H_I = \sum_{\alpha=\{x,y,z\}} \sum_{i=1}^{n} c_\alpha s_\alpha^{(i)} \otimes b_\alpha = \sum_{\alpha=\{x,y,z\}} S_\alpha^n \otimes B_\alpha$$

$s_\alpha^{(i)}$: system operator acting on ith qubit

$b_{\alpha,i}$: bath operator

$c_{\alpha,i} = c_\alpha$

$$s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I$$

$\lambda \gg s$

collective spin operators: S_α^n

bath couples identically to each qubit

$$\mathbf{S}_\alpha |\psi\rangle = 0$$

$$\mathbf{H}_I |\psi\rangle \otimes |\phi_{env}\rangle = 0$$

for $|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$

Collective errors do not err the decoherence-free subspace

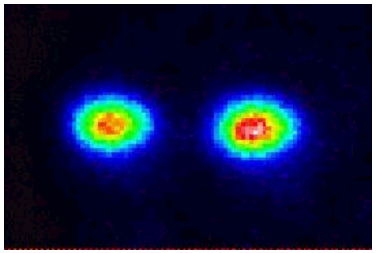
Decoherence-Free In Practice

SCIENCE VOL 291 9 FEBRUARY 2001

REPORTS

A Decoherence-Free Quantum Memory Using Trapped Ions

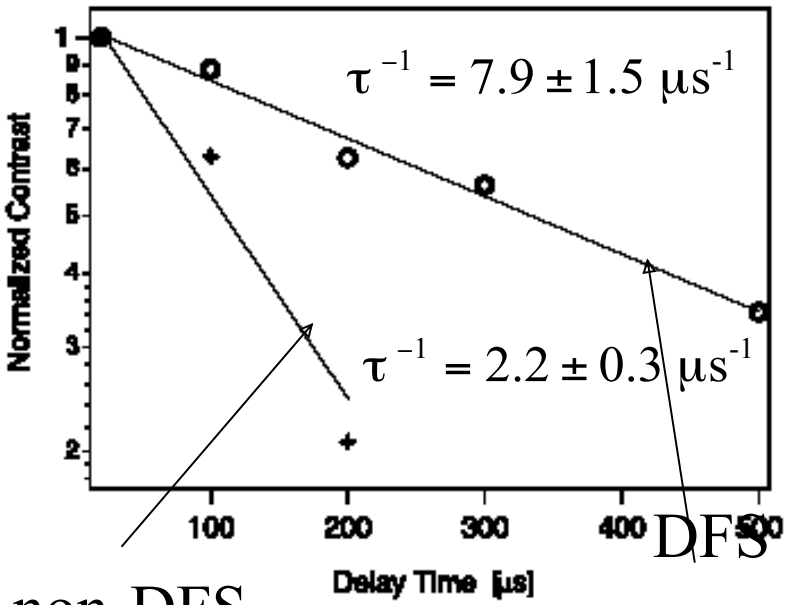
D. Kielpinski,[1]* V. Meyer,[1] M. A. Rowe,[1] C. A. Sackett,[1]
W. M. Itano,[1] C. Monroe,[2] D. J. Wineland[1]



collective phase errors:

$$S_z^n = \sum_{i=1}^{n} s_z^{(i)}$$

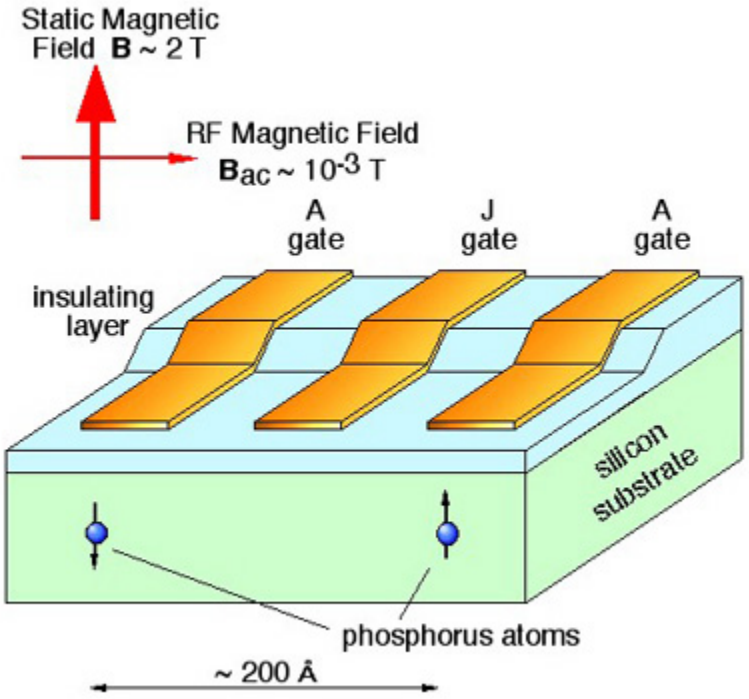
ambient conditions!



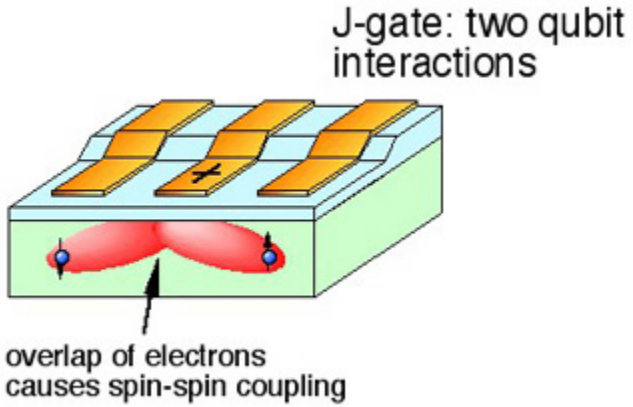
DFS

non-DFS

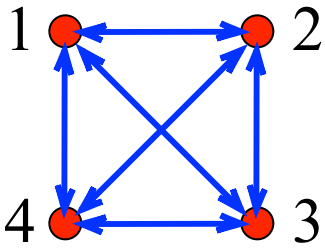
Spintronic Supercoherent



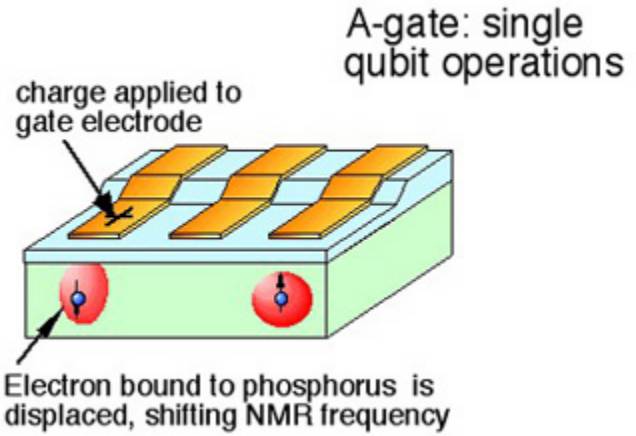
(Kane model)
Kane, Nature 393, 133
(



~0.1 meV exchange interaction

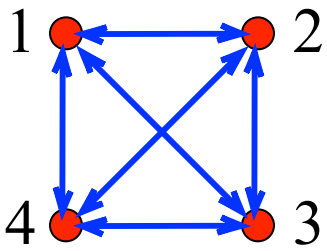


T<1K supercoherence



diagrams: Los Alamos Quantum

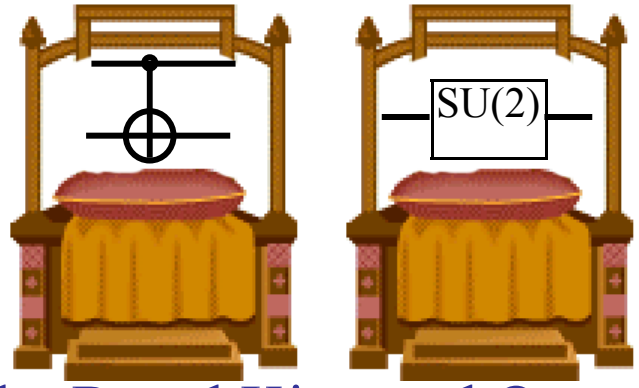
The Full Computer



= 1 (encoded) qubit

?

full quantum architecture



The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Universal Quantum Computer

•a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

•a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.

$$\text{SU(2)} = \cos(\theta)\mathbf{I} + i\sin(\theta)\hat{n}\cdot\hat{\boldsymbol{\sigma}}$$

$$\equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$$

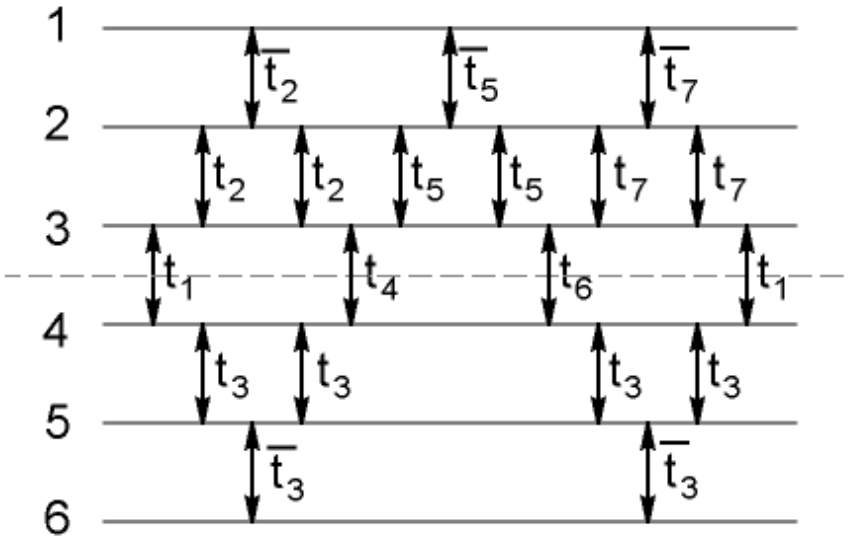
Encoded Universality

[Bacon, Kempe, Lidar, and Whaley, *Phys. Rev. Lett.* 85, 1758 (2000)]
[Bacon, Kempe, DiVincenzo, Lidar, and Whaley, *Proc. of the Int. Conf. on Exp. Imp. of Quant. Comp., Sydney, Australia* (2001)]
[DiVincenzo, Bacon, Kempe, Burkard, and Whaley, *Nature* 408, 339 (2001)]
[Bremner, Dodd, Nielsen, and Bacon, *Phys. Rev. A* 69 012313 (2004)]

Exchange cannot implement all unitary operations

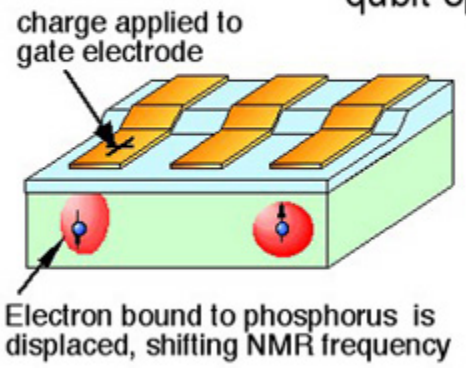
$$[\mathrm{H}_{ex}, \mathrm{S}_{\alpha}] = 0$$

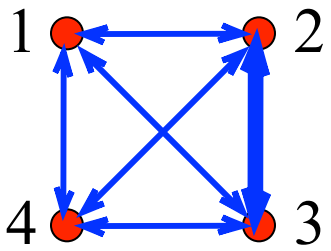
If we use encoded subspaces exchange is universal!



t_1=0.410899(2) t_5=0.414720(10)
t_2=0.207110(20) t_6=0.147654(12)
t_3=0.2775258(12) t_7=0.813126(12)
t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

encoded controlled-not





Use Physics!

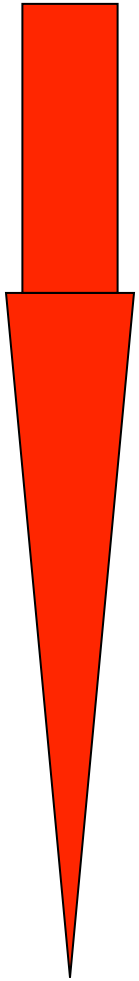
Supercoherent Qubits

Quantum Error Detecting Codes

Decoherence-Free Subspaces

Solid State Implementation

Encoded Universality



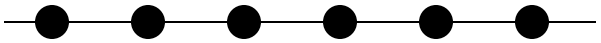
Supercoherent Quantum Computing Architecture

The Physics Guarantee

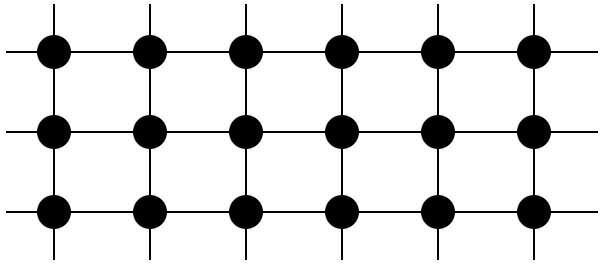
Classical Ising memory

$$\mathbf{H} = -J \sum_{\langle i,j \rangle} \boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$$

One dimensional Ising model



Two dimensional Ising model

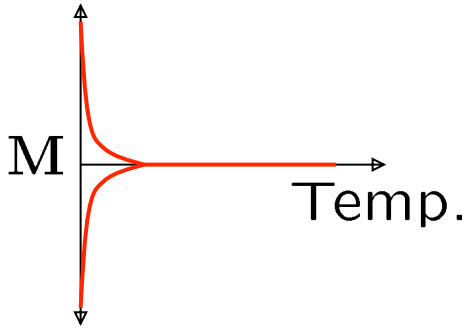


Redundancy code

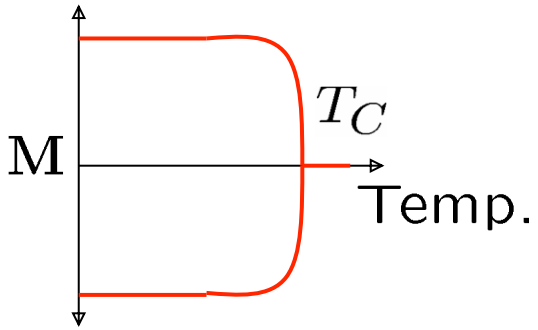
Codewords are ground states $|0 \cdots 0\rangle$, $|1 \cdots 1\rangle$

Order parameter: total magnetization, $\mathbf{M} = \sum_i \boldsymbol{\sigma}_z^{(i)}$

indication that the information is "still there"

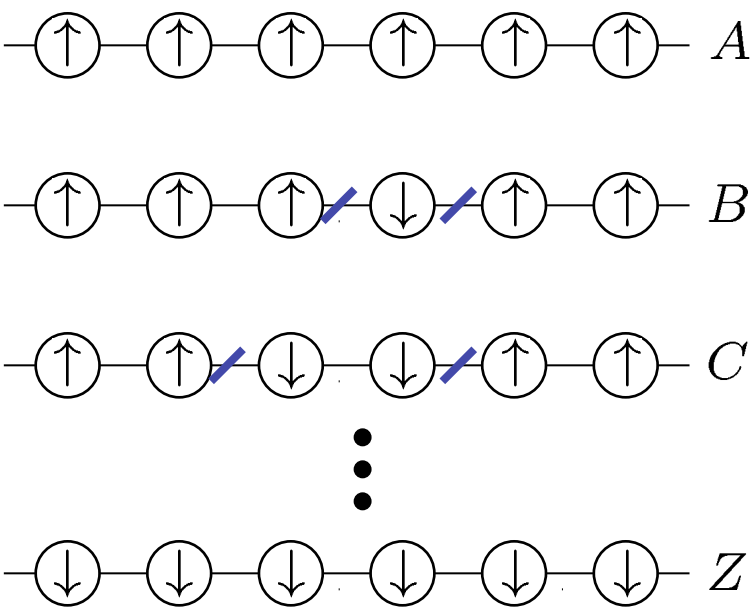


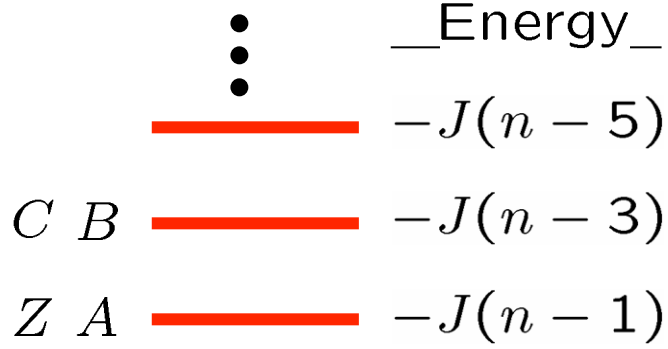
Don't build memory in 1D



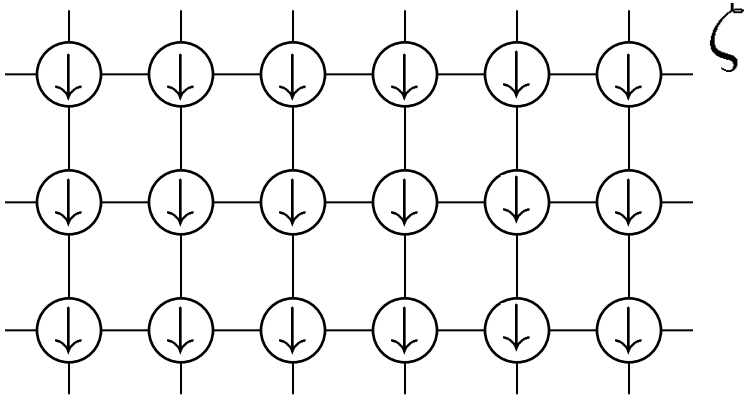
2D memories much better

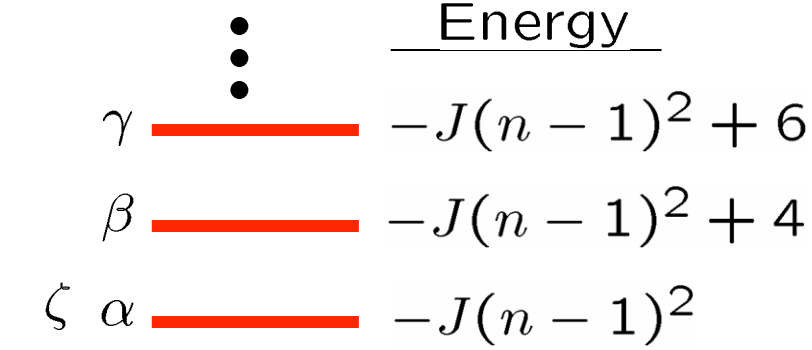
Self Fixing





Excitations pointlike.
Energy cost constant.
disordering entropy $\gg$ energy



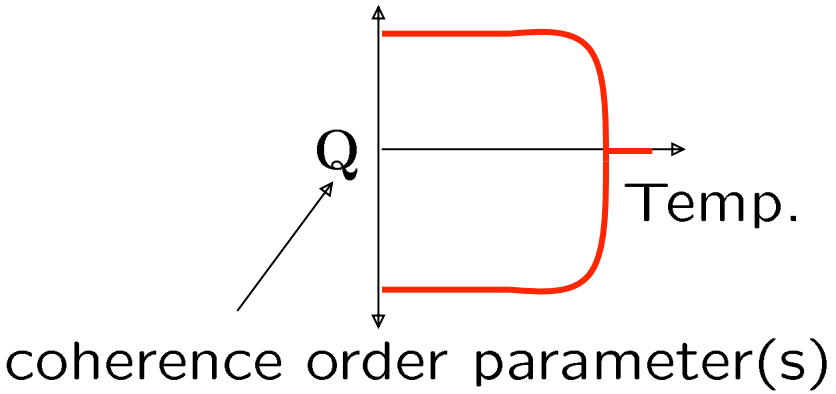


Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

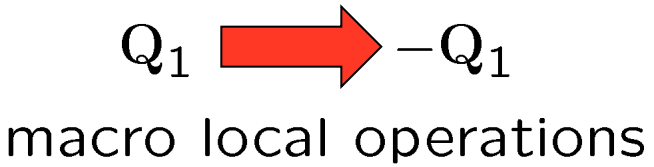
The Quantum Hard Drive?

[Bacon, Ph.D. Thesis, U.C. Berkeley Physics "Decoherence, Control, and Symmetry in Quantum Computers," (2001)]

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.



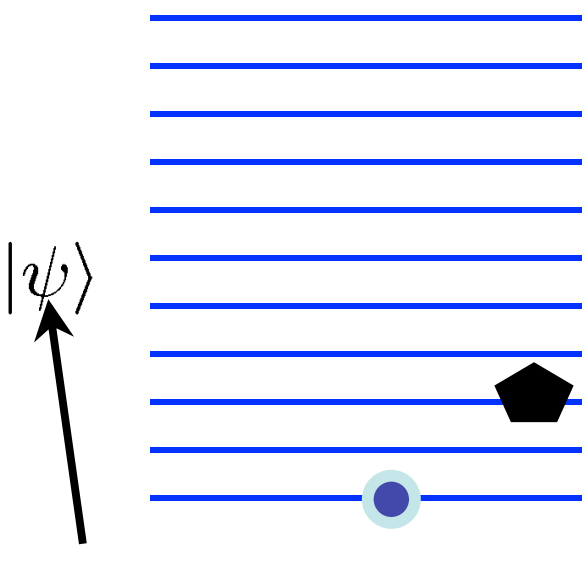
2. Accessible Fault-Tolerant Operations

"self-correcting"

$$\left\langle G_1, \ldots, G_{n^k} \right\rangle \cong U(c^n)$$

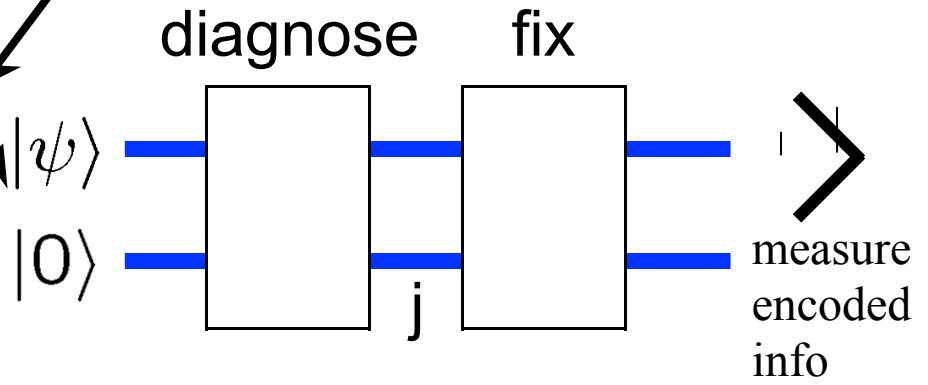
3. Universality

Quantum Error Correcting Order Parameter



ρ_{error}

The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

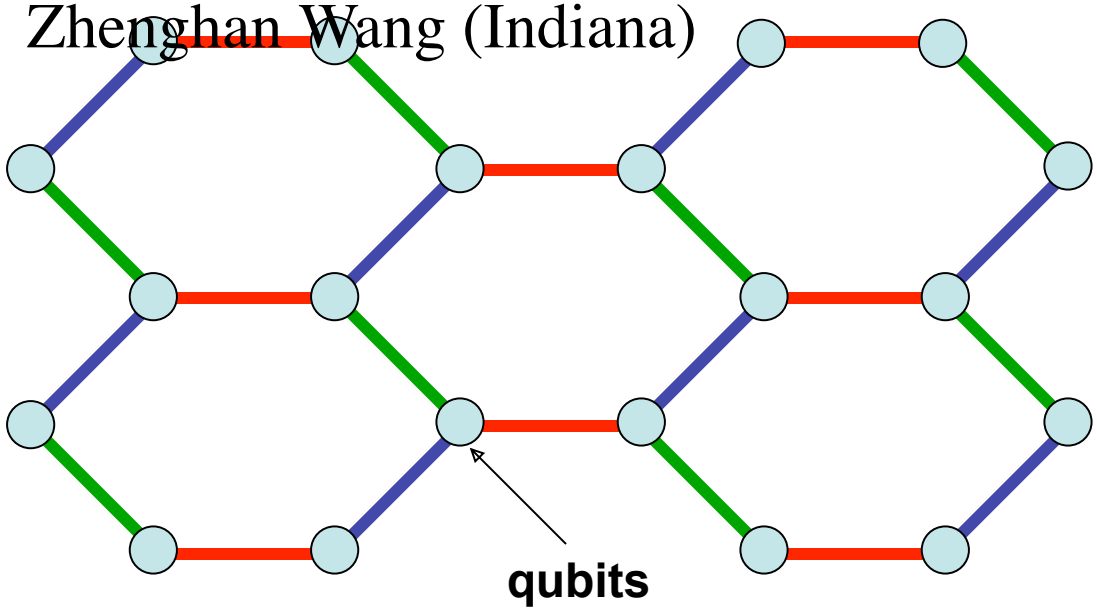
$\mathbf{L}_s$ logical encoded operation

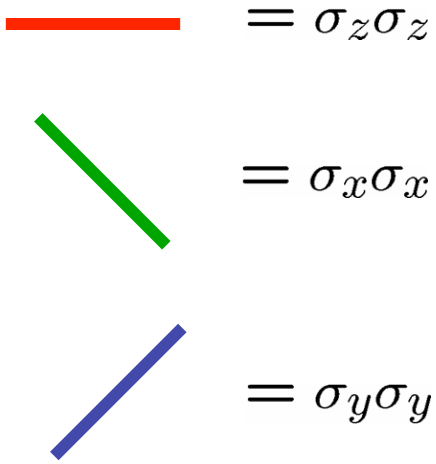
$\mathbf{D}_j$ projector onto jth errored subspace

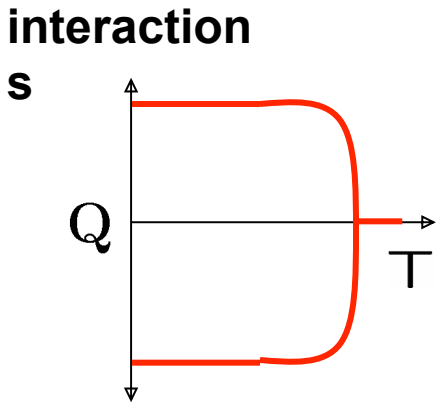
$\mathbf{F}_j$ fixing operator for jth errored subspace

Quantum Computing Phase

Alexei Kitaev (Caltech)
Michael Freedman (Microsoft)
John Preskill (Caltech)
Zhenghan Wang (Indiana)
Lev Ioffe (Rutgers)
Michael Larsen (Indiana)
Chetan Nayak (UCLA)
Fei Zhou (UBC)





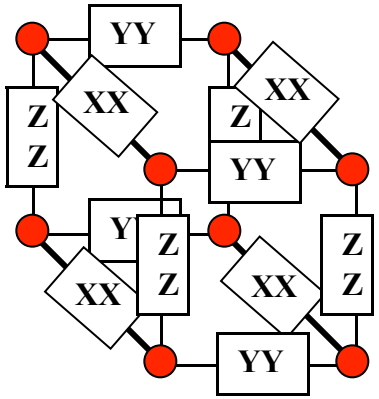


Lacking: Self-correcting (natural fault-tolerance)

(exception 4D toric code of Preskill et. al.)

Into the Third Dimension

$n \times n \times n$



Cubic lattice with qubits at vertices

$$\mathbf{X}_{i,j,k} = \boldsymbol{\sigma} x|_{i,j,k} \text{ th qubit}$$
$$\mathbf{Y}_{i,j,k} = \boldsymbol{\sigma} y|_{i,j,k} \text{ th qubit}$$
$$\mathbf{Z}_{i,j,k} = \boldsymbol{\sigma} x|_{i,j,k} \text{ th qubit}$$

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

Ground state = quantum error correcting codespace
encodes 1 qubit, can correct up to n errors

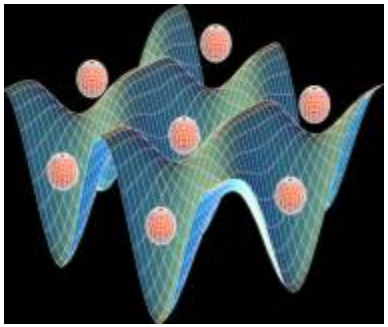
Numerical evidence for self-correction (string-like excitations)

Does this system exhibit “natural fault-tolerance?”

Optical lattice as natural fault tolerant testbed:

Demler and Zhou, *Phys. Rev. Lett.* 88, 163001(2002).

Duan, Demler, and Lukin, *Phys. Rev. Lett.* 91, 090402 (2003).



Error correcting codes

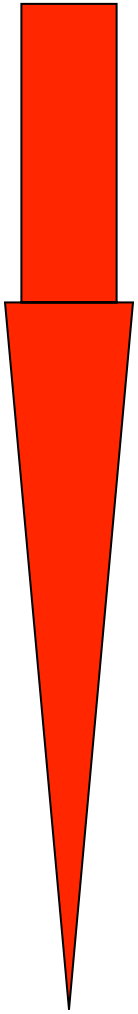
- classical
- quantum

Fault-tolerance

- local structures giving rise to global robustness

Representation theory

- identification of qubits and logical operations



Geometry, Topology?

Statistical Physics

- theory of strongly interacting quantum systems
- renormalization group methods
- new order parameters?

Implementation Physics

- let physics guide
- optical lattices, donor electrons, superconducting circuits, ion traps, etc.

Naturally Fault-Tolerant Quantum Computation

But WHAT do we do when we get there?

Shor's Algorithm (1994)



Peter Shor

188198812920607963838697239461650439807163563379417382700763356

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Quantum computer's can efficiently factor numbers (and hence break RSA cryptosystems)

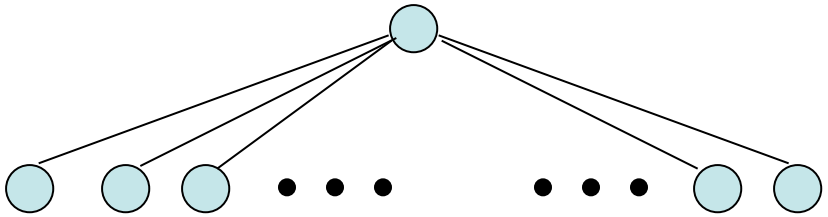
(It is widely suspected that classical computers cannot efficiently factor numbers.)



Where are the Quantum Algorithms?

Quantum Power?

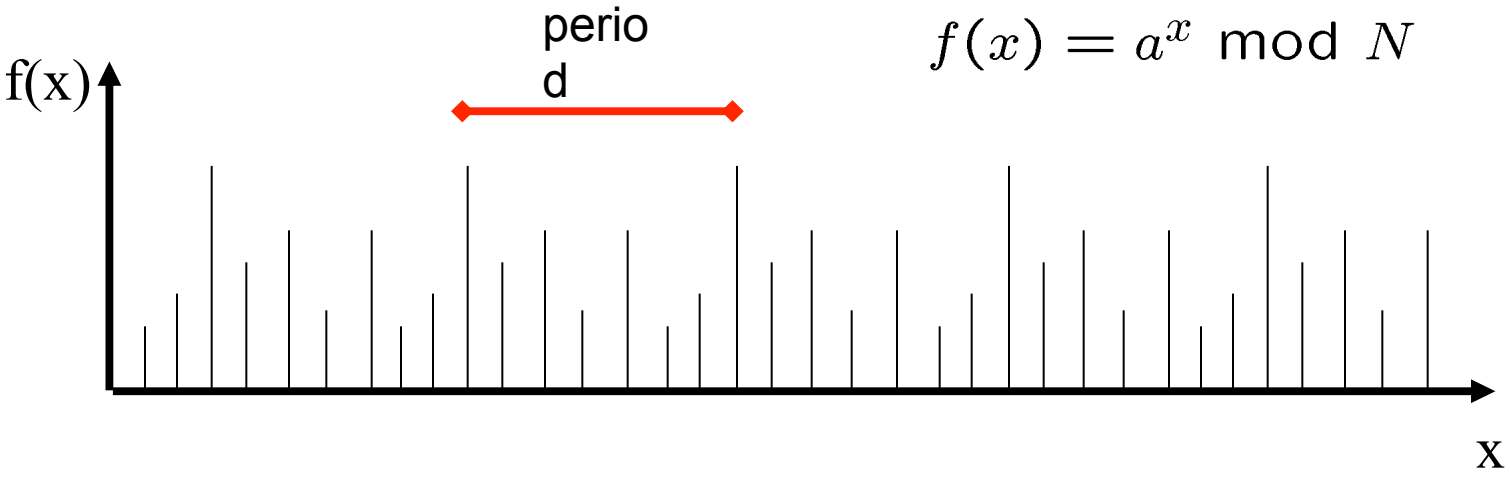
$$|\psi_{QC}\rangle = \alpha_1|\text{comp}_1\rangle + \alpha_2|\text{comp}_2\rangle + \cdots + \alpha_{2^n}|\text{comp}_{2^n}\rangle$$



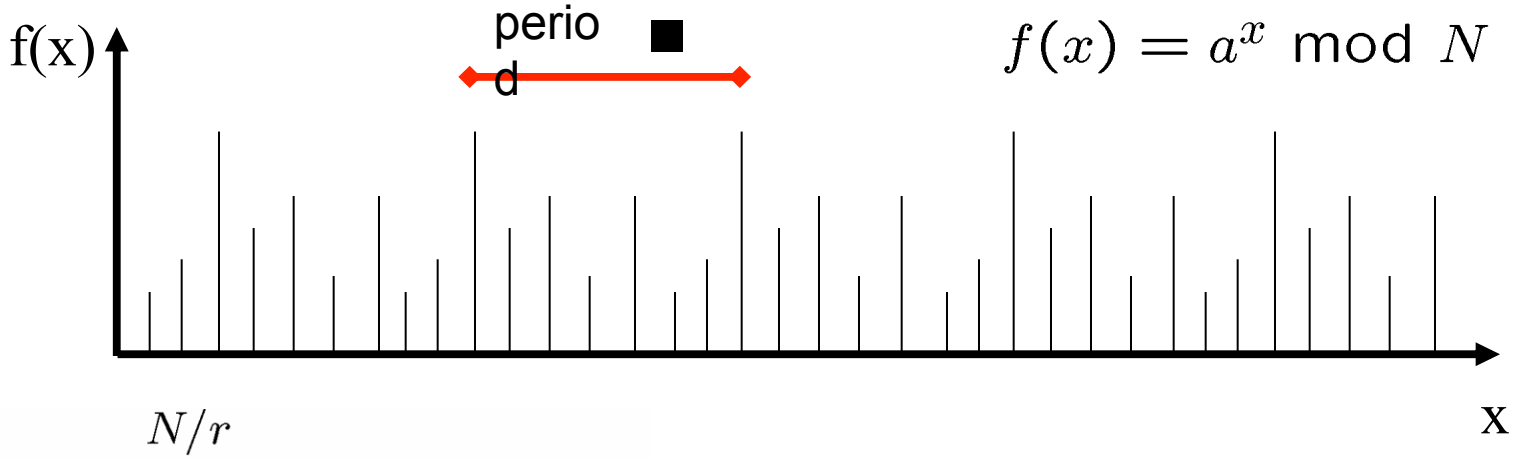
Quantum Parallelism? **NOT REALLY**

Quantum computers don't "check all possible factors"

Shor's insight: Period finding can be used to factor numbers



Symmetry



$$|\psi_c\rangle = \sum_{i=1}^{N/r} |ir + c \bmod N\rangle \qquad c = 0, 1, \ldots, N/r - 1$$

$$\boldsymbol{\rho} = \frac{1}{N/r} \sum_c |\psi_c\rangle\langle\psi_c|$$

$\boldsymbol{\rho}$ is invariant under the abelian group of displacements

$$\mathbf{T}^k \boldsymbol{\rho} \mathbf{T}^{-k} = \boldsymbol{\rho} \qquad \mathbf{T}|x\rangle = |x + 1\rangle$$

invariance under displacement $\Rightarrow$ work in (discrete) momentum basis

Symmetries $\Rightarrow$ change of basis

Negative Probabilities

Classical :

$$\boldsymbol{\rho} = \sum_i p_i |i\rangle\langle i|$$

$$\boldsymbol{\rho} \to \sum_s q_s \mathbf{M}_s \boldsymbol{\rho} \mathbf{M}_s^{-1}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$q_s > 0,\ \sum_s q_s = 1,\ \mathbf{M}$ permutation matrix

$$|p\rangle = \frac{1}{\sqrt{N}} \sum_x \exp\left(\frac{2\pi i x p}{N}\right) |x\rangle$$

some transitions elements must be negative

Classically we <u>cannot</u> achieve basis change of an arbitrary symmetry

Quantum:

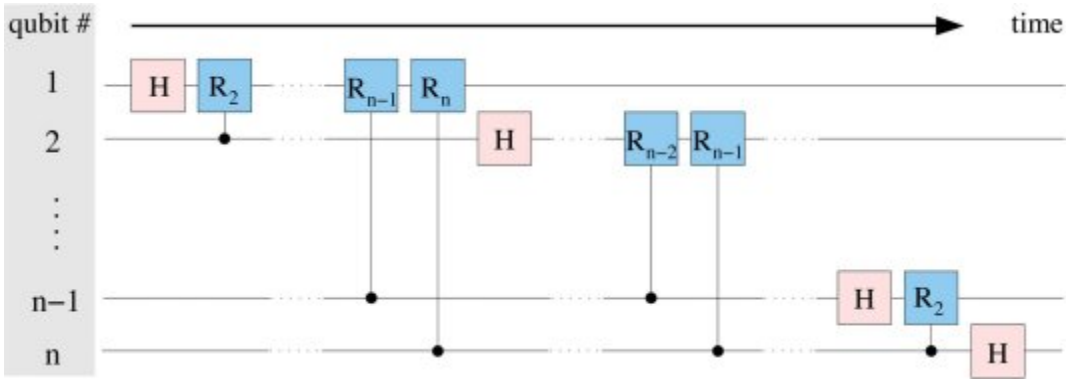
$$\boldsymbol{\rho} \to \mathbf{U} \boldsymbol{\rho} \mathbf{U}^\dagger$$

Quantum we <u>can</u> (almost always) change basis for arbitrary symmetry.

Quantum computers can exploit symmetries of problem which a classical computer cannot exploit.

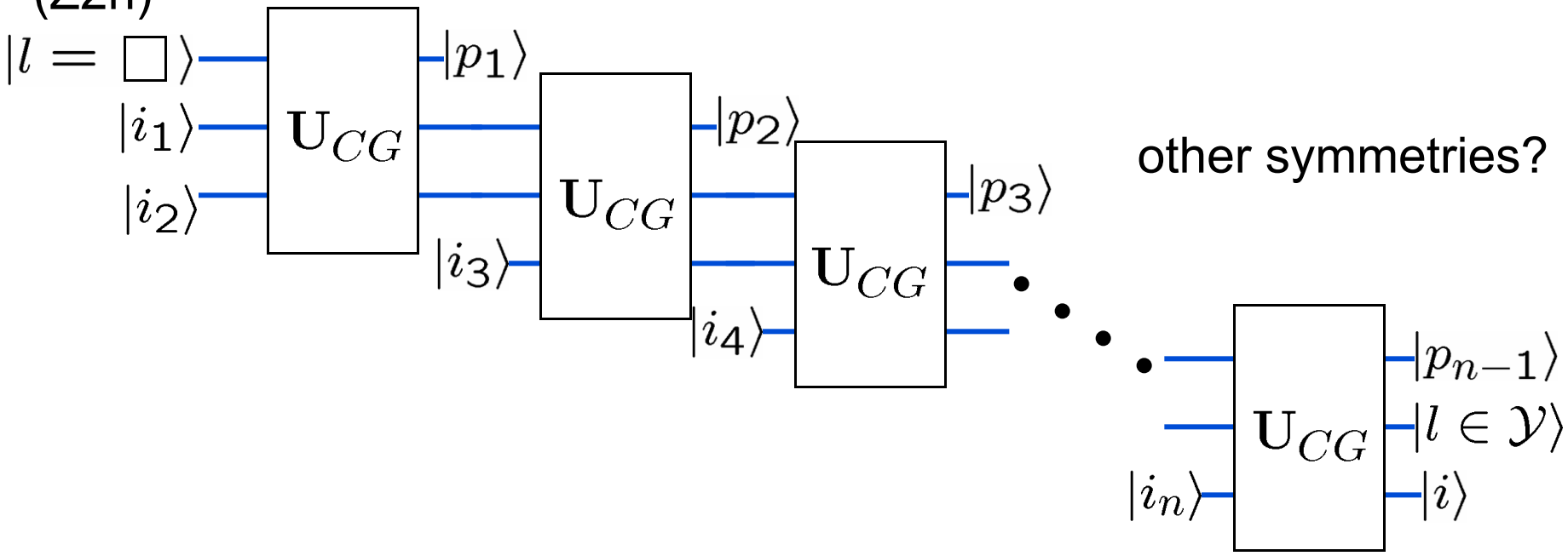
see for example Schützhold and Unruh, quant-ph/0304090

Efficient Circuits



displacement symmetry

Quantum Circuit for the Fourier Transform (Z2n)



other symmetries?

Schur Duality Transform

[Bacon, Chuang, and Harrow, in preparation (2004)]

n identical copies of a state:

$$\rho \otimes \rho \otimes \cdots \otimes \rho$$

permutation symmetry

independent of basis: $\int dU \mathbf{U}^{\otimes n} \rho \mathbf{U}^{\dagger \otimes n}$

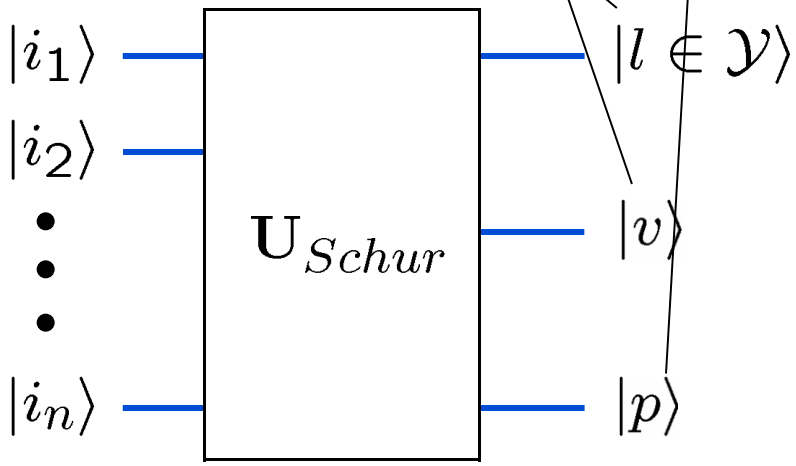
unitary symmetry

$$\mathbf{P}_\pi = \bigoplus_{l \in \mathcal{Y}} \mathbf{I}_{n_l} \otimes \mathbf{S}_\pi^l$$

$\mathbf{S}_\pi^l =$ irrep of $\mathcal{S}_n$

$$\mathbf{U}^{\otimes n} = \bigoplus_{l \in \mathcal{Y}} \mathbf{V}_\mathbf{U}^l \otimes \mathbf{I}_{d_l}$$

$\mathbf{V}_\mathbf{U}^l =$ irrep of $\mathcal{U}(d)$



$\mathcal{Y} =$ set of all partitions of n
(all Young Diagrams with n rows)

$$l = (l_1, l_2, \ldots, l_n) \qquad \sum_{i=1}^{n} l_i = n$$
$$l_1 \geq l_2 \geq \ldots \geq l_n \geq 0$$

Our result: An efficient quantum circuit for Schur Duality Transform

Applications

Spectrum Estimation: **[Keyl and Werner, Phys. Rev. A 64 052311 (2001)]**

$\rho \otimes \rho \otimes \cdots \otimes \rho$

Task: estimate eigenvalues of ρ

Perform $\mathbf{U}_{Schur}$

Measure $|l \in \mathcal{Y}\rangle$

Estimate $\left(\frac{l_1}{n}, \frac{l_2}{n}, \ldots, \frac{l_n}{n}\right)$

Quantum Universal Variable Length Source Coding

[Hayashi and Matsumoto, quant-ph/0209124]

Quantum Hypothesis Testing

[Hayashi, quant-ph/0107004]

State Estimation

[Gill and Masar, quant-ph/9902063]

Encoding Into Decoherence-Free Subspaces

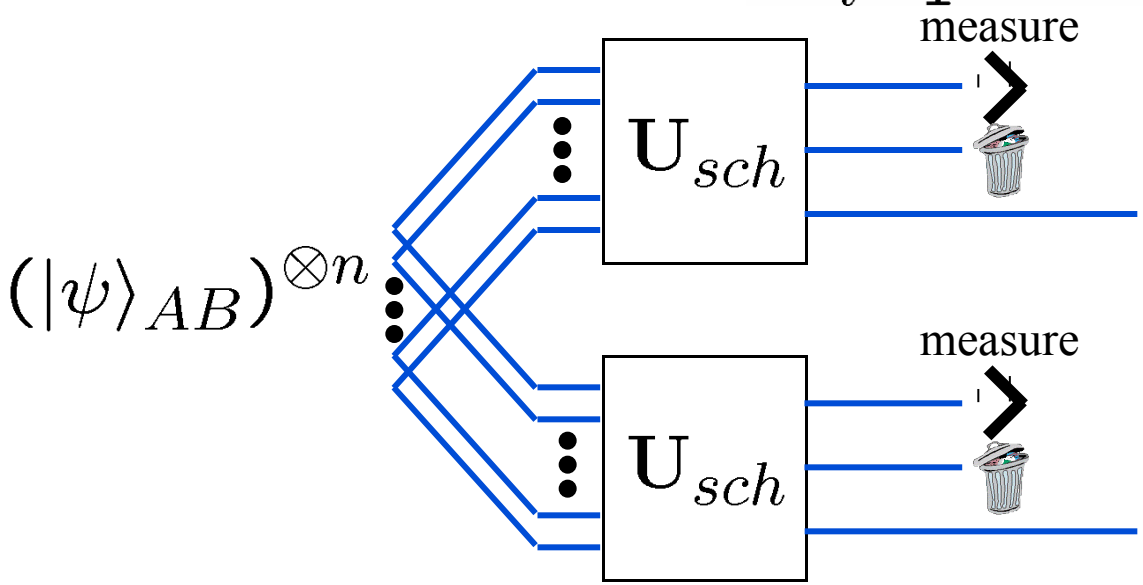
[Bacon, Kempe, Lidar, and Whaley, *Phys. Rev. Lett.* 85, 1758 (2000)]
[Kempe, Bacon, Lidar, and Whaley, *Phys. Rev. A* 63, 042307 (2001)]

Universal Entanglement Concentration

[Hayashi and Matsumoto, quant-ph/0209030]

$(|\psi\rangle_{AB})^{\otimes n}$ $|\psi\rangle_{AB} =$ unknown entangled state

$$= \sum_{i=1}^{d} c_i |i\rangle_A \otimes |i\rangle_B$$



$$\frac{1}{\sqrt{d_e}} \sum_{i=1}^{d_e} |i\rangle_A |i\rangle_B$$

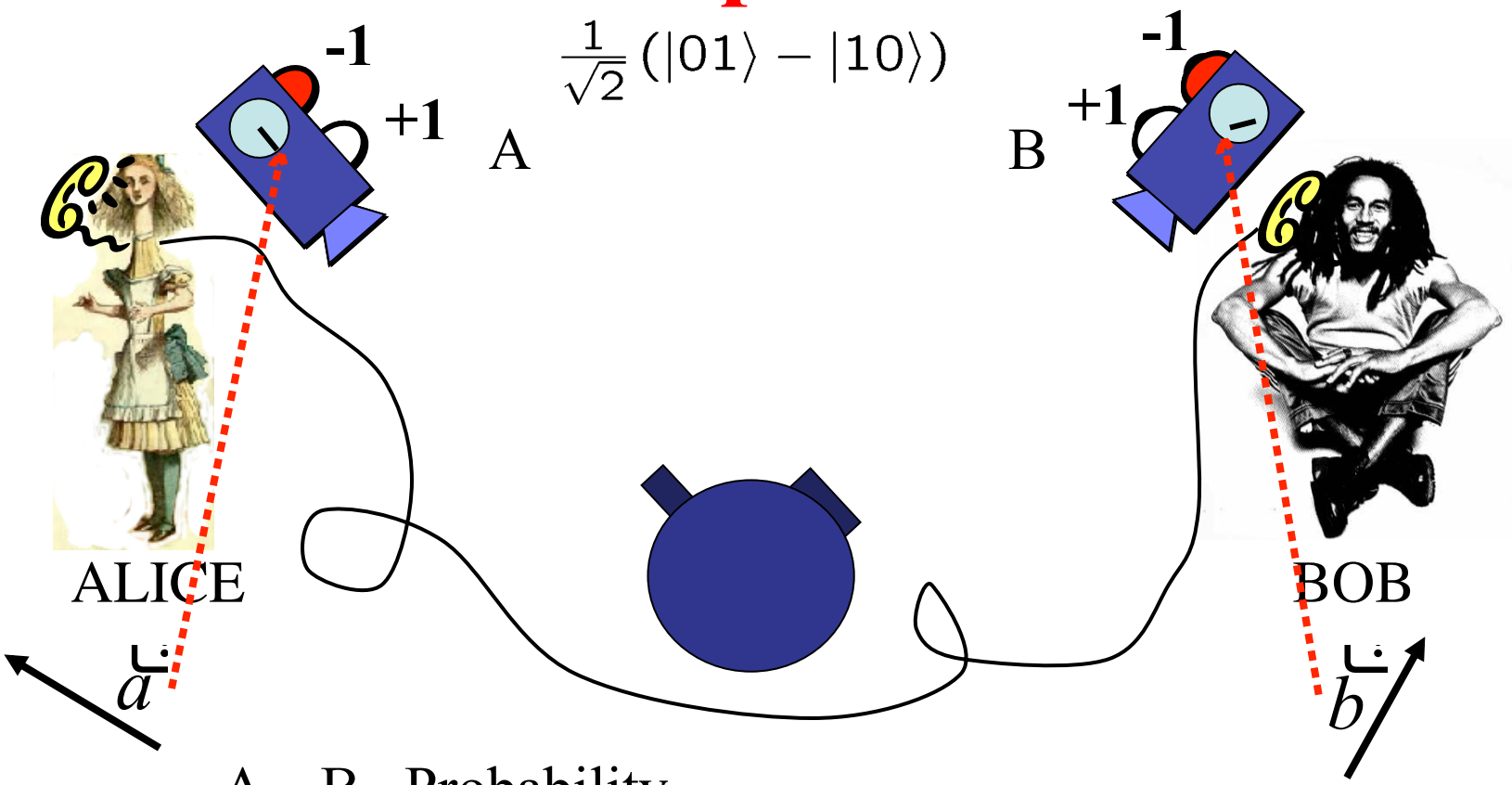
$$d_e \approx 2^{nE(|\psi\rangle\langle\psi|)}$$

$$E(|\psi\rangle\langle\psi|) = -\mathrm{Tr}[\boldsymbol{\rho}_A \ln \boldsymbol{\rho}_A]$$

Von Neumann entropy

Maximally entangled states are the “currency” of bipartite entanglement.

Bell Experiment



A	B	Probability
- 1	- 1	$\frac{1}{4}\left(1-\vec{a}\cdot\vec{b}\right)$
1	+1	$\frac{1}{4}\left(1+\vec{a}\cdot\vec{b}\right)$
+1	- 1	$\frac{1}{4}\left(1+\vec{a}\cdot\vec{b}\right)$
+1	+1	$\frac{1}{4}\left(1-\vec{a}\cdot\vec{b}\right)$

$$\langle A\rangle = \langle B\rangle = 0$$

$$\langle AB\rangle = -\vec{a}\cdot\vec{b}$$

A Conversation

$$\Pr(A, B|\hat{a}, \hat{b}) = \frac{1}{4}\left(1 - AB\hat{a}\cdot\hat{b}\right)$$

Computer Scientist: “How are these correlations different than shared random variables?”

Physicist: “John Bell showed that these correlations cannot be created using shared random variables. We physicists say `there is no local hidden variable theory for these correlations.’”

Computer Scientist: “OK. So how much communication in addition to shared random variables is needed in order to produce these correlations?”

Physicist: “What a silly question! This communication would have to be superluminal signals.”

Computer Scientist: “But I only care about comparing resources.”

[Brassard, Cleve, and Tapp, *Phys. Rev. Lett* 83, 1874 (1999)]

Why Is It Hard?

$$\Pr(A, B|\hat{a}, \hat{b}) = \frac{1}{4}\left(1 - AB\hat{a}\cdot\hat{b}\right)$$

How to produce distribution without sending $\hat{a}$ from A to B (or vice versa)?

requires infinite number of bits



ALICE

Stop running up my phone bill, Bob!



BOB

Simulating Quantum Correlations

[Masar, Bacon, Cleve, and Cerf, *Phys. Rev. A* 63, 052305 (2001)]
[Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]

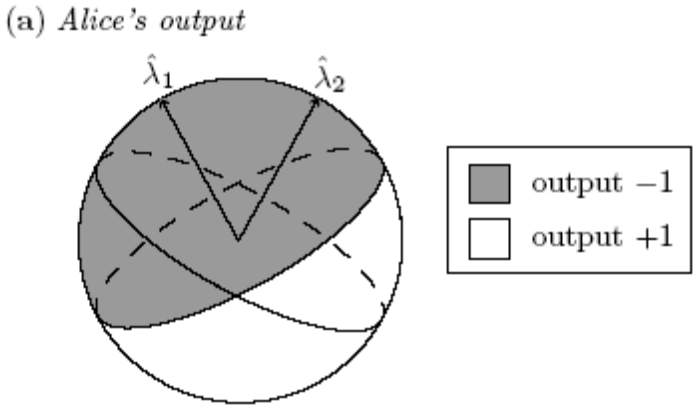
correlations from projective measurements on Bell pair

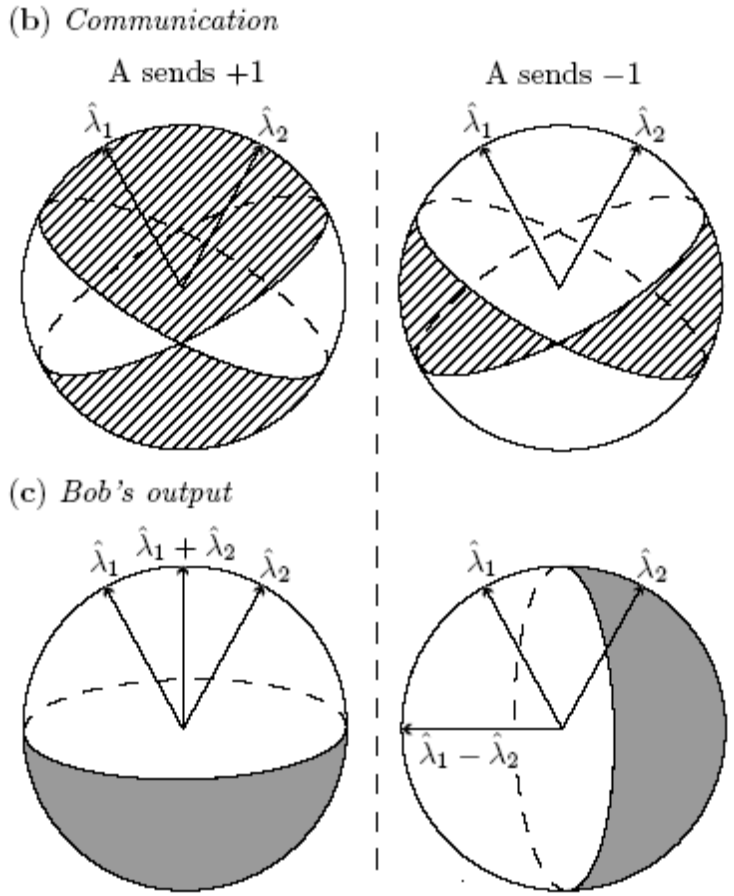
$=$

shared randomness

$+$

1 bit of communication





Quantum Communication Complexity



x

ALICE

F(x,y)

How much (how many bits) do they need to communicate to compute F?



y

BOB



Ran Raz

If they share entangled quantum states, then there are F(x,y)'s such that A and B need exponentially less communication than if they did not share entanglement.

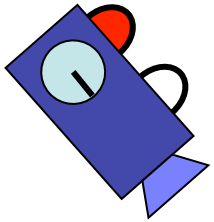
[Raz, Proceeding of the 31st STOC, 358 (1999)]

The communication cost of simulating quantum correlations!

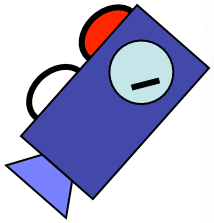
Detector Inefficiency



[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]



$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$





Bell experiment

ALICE

$$|\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle| \leq 2$$

BOB

If detectors sometimes do not fire, then there may be a local hidden variable explanation of the experiment = detector inefficiency loophole

correlations from projective measurements on Bell pair

=

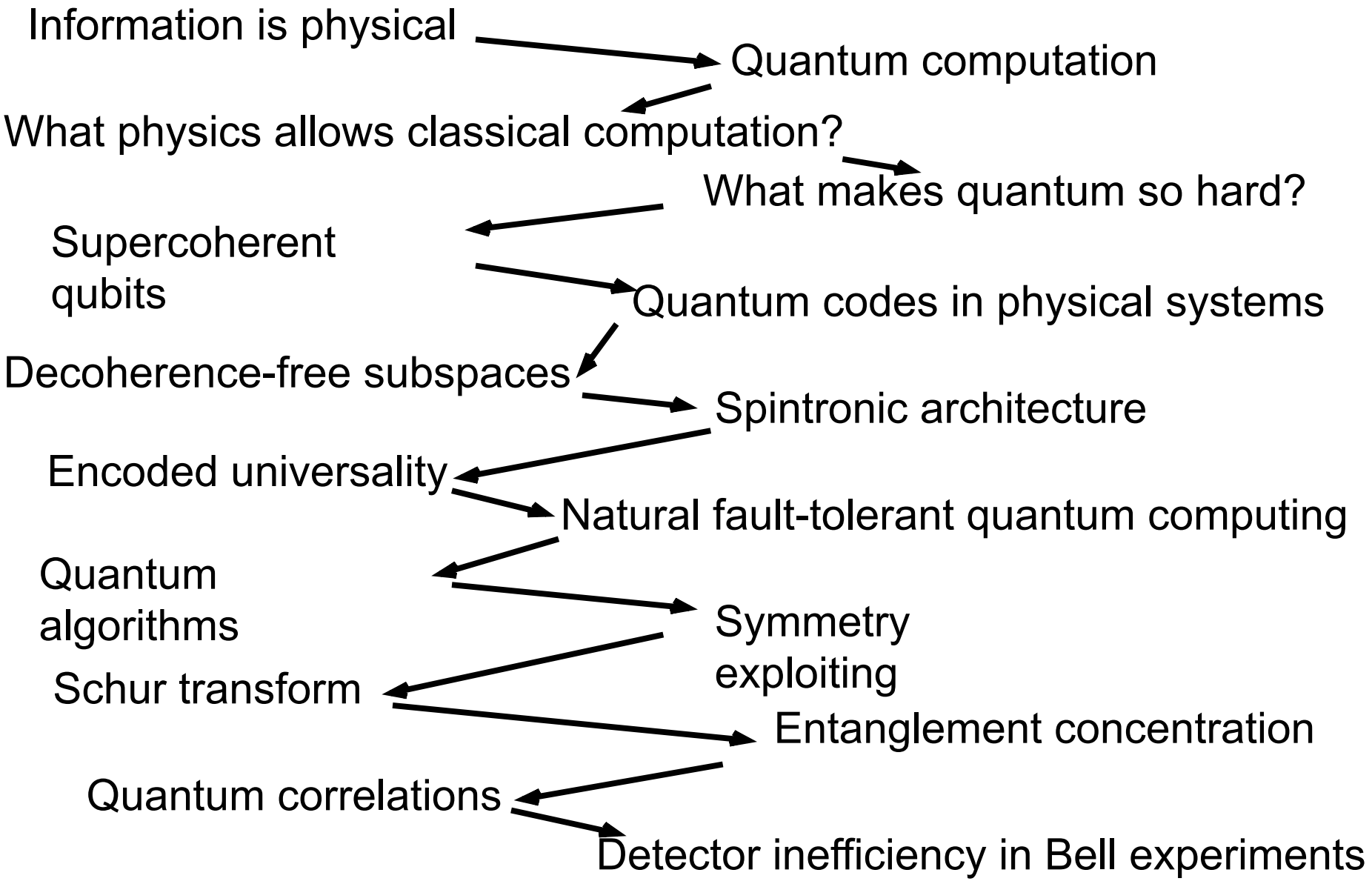
shared randomness + 1 bit of communication

Example:
efficiency<82.8% = LHV

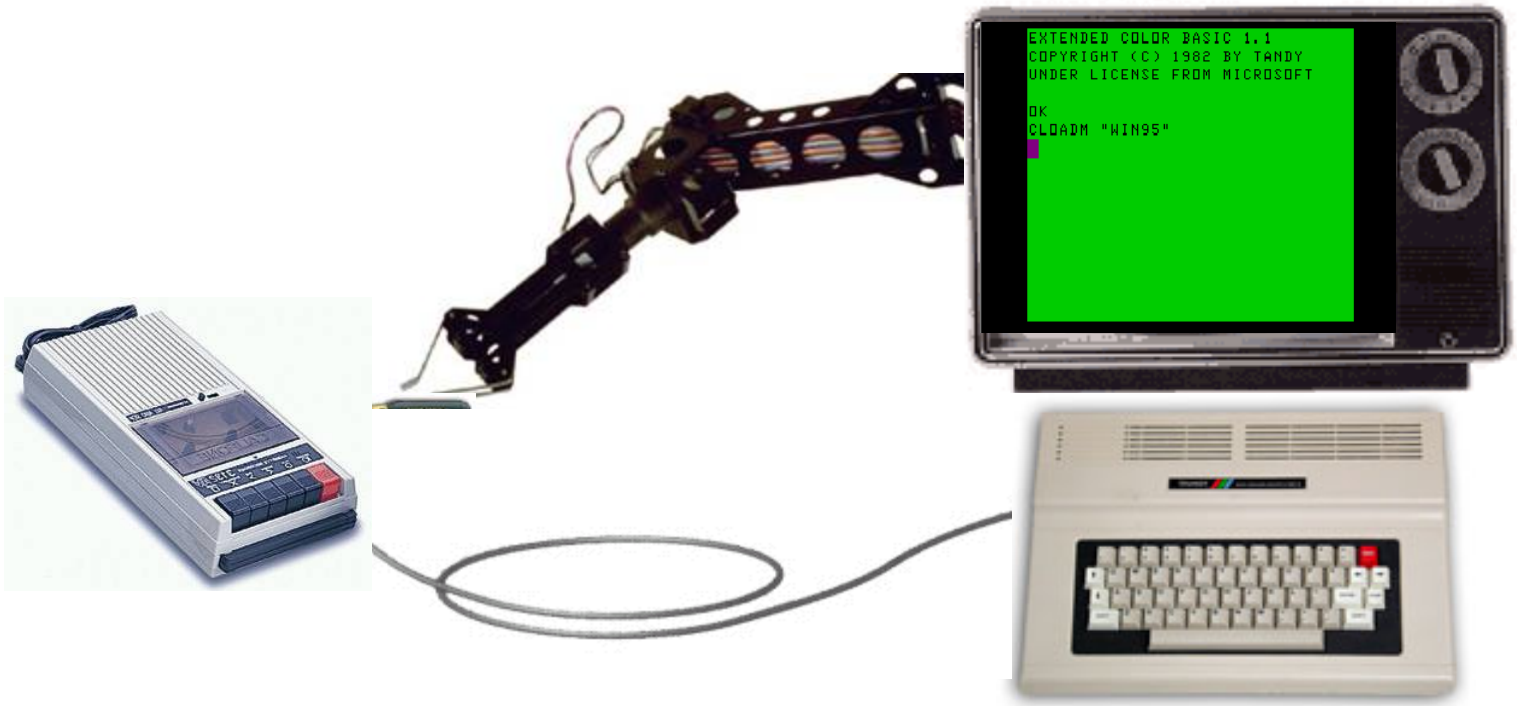
Limit on detector inefficiency!

Theoretical Computer Science Giving Back to Physics!

Our Path


Information is physical
Quantum computation
What physics allows classical computation?
What makes quantum so hard?
Supercoherent qubits
Quantum codes in physical systems
Decoherence-free subspaces
Spintronic architecture
Encoded universality
Natural fault-tolerant quantum computing
Quantum algorithms
Symmetry exploiting
Schur transform
Entanglement concentration
Quantum correlations
Detector inefficiency in Bell experiments

The Path Ahead (I)



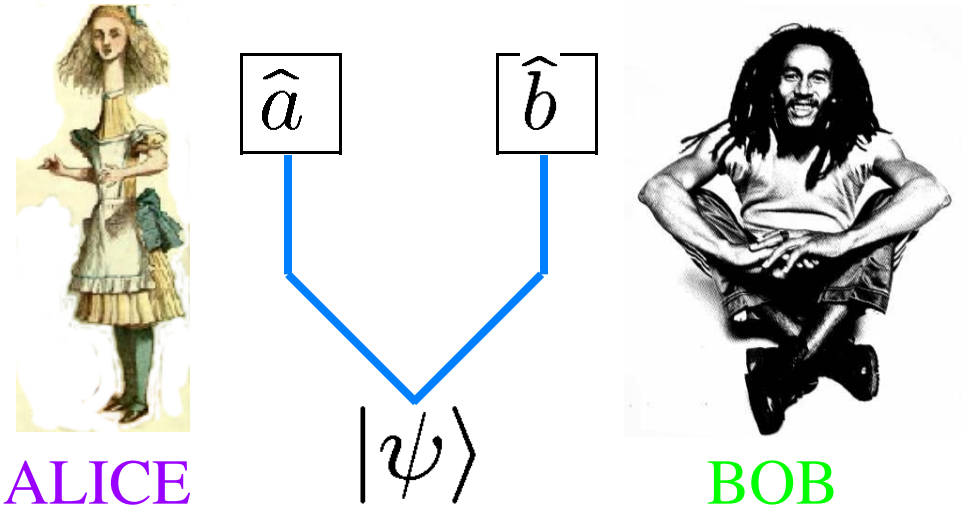
Quantum Computers that Fix Themselves

The Path Ahead (II)



Quantum Algorithm's and Symmetry

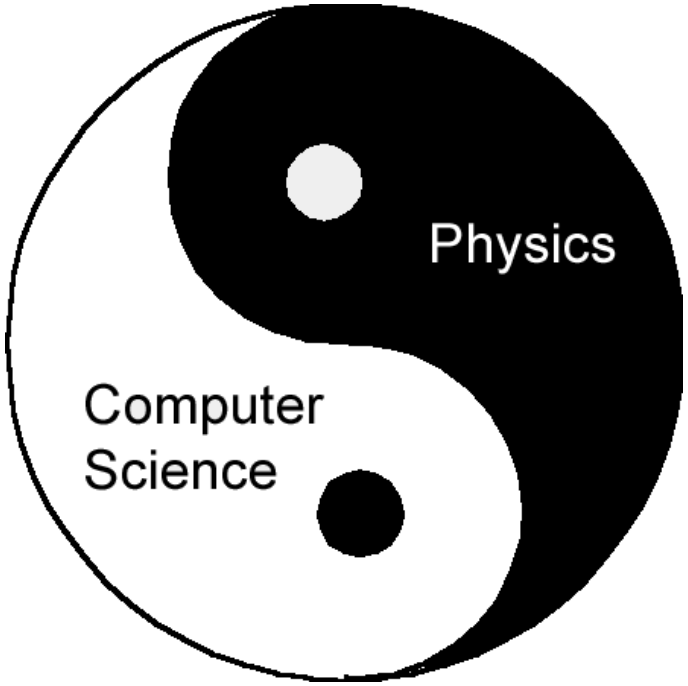
The Path Ahead (III)



$d_A d_B$ dimensional state pure state

Towards a Theory of Quantum Correlations

Bigger Picture



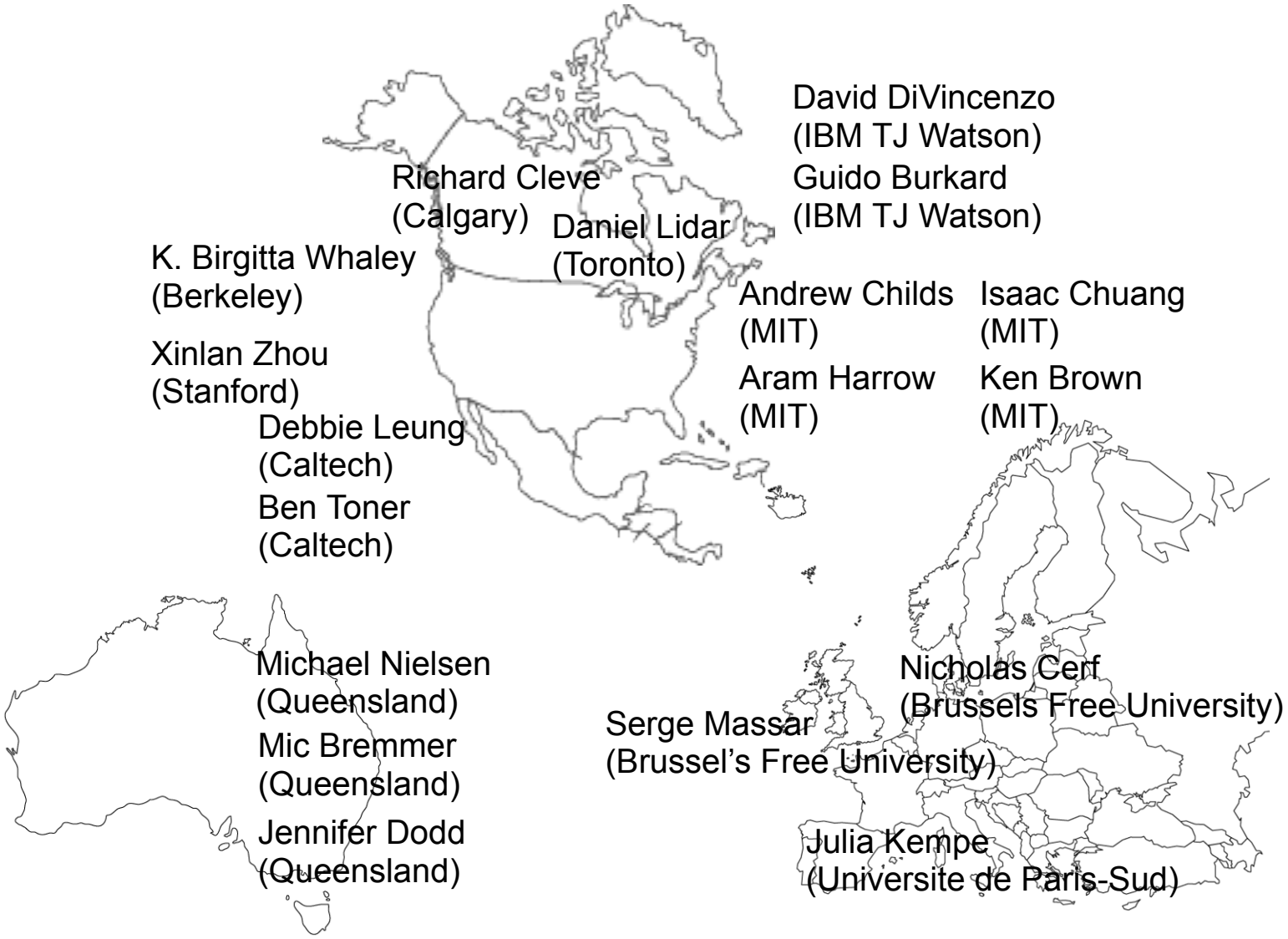
Quantum computation

building
exploiting
understanding

Quantum computation

what can we learn about physics from theoretical computer science?

Collaboratorium



Fin

